FOUR EXPONENTIALS AT CM PERIODS, AND A BILINEAR STATEMENT FOR TWO CM ELLIPTIC CURVES

DRAFT — NOT FOR CIRCULATION

Abstract. Let Ω be a nonzero period of an elliptic curve with complex multiplication, defined over $\overline{\mathbb{Q}}$. We prove the four exponentials conjecture at $t = c\,\Omega$ and at $t = c\,\Omega/\pi$ (and hence at their inverses) for every $c \in \overline{\mathbb{Q}}^{\times}$: for instance, 2^{ϖ} and 3^{ϖ} are not both algebraic, and neither are $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$, where ϖ is the real period of $y^2 = 4x^3 - 4x$. We also show that one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ is transcendental. These follow from dimension bounds of four-exponentials shape on $\mathbb{G}_{\mathrm{m}}^2 \times E$ and $\mathbb{G}_{\mathrm{m}}^2 \times E^{\natural}$. A second application of the same mechanism, to two non-isogenous CM curves, gives a bilinear statement: if u is a non-torsion elliptic logarithm on $y^2 = 4x^3 - 4x$, Ω_ρ a period of $y^2 = 4x^3 - 4$, and a, t are algebraic, then $\varpi\,\Omega_\rho\,(\zeta(u)+t)/(\pi(u+a))$ is transcendental. The proofs of these results apply Waldschmidt's algebraic subgroup theorem with period term (1988), in the K-version of Roy and Waldschmidt, directly to explicit quotient groups whose connected algebraic subgroups we list. The period term comes from a period of $\mathbb{G}_{\mathrm{m}} \times E$ (resp. $\mathbb{G}_{\mathrm{m}} \times E^{\natural}$) lying on the relevant line, and in the two-curve case from the Legendre relations of both curves. We state precisely what the proofs use from the literature, and write out the one step of Waldschmidt's auxiliary-function construction (multiplicities along a subspace) that is only indicated in print. We also record a single-curve theorem ("Theorem M") with a second proof by Laurent's identity and Wüstholz's analytic subgroup theorem.

1. Introduction

Let $\overline{\mathbb{Q}}$ be the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, $\mathcal{L} = \{z \in \mathbb{C} : e^z \in \overline{\mathbb{Q}}^{\times}\}$ and $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. For an elliptic curve $E\colon y^2 = 4x^3 - g_2x - g_3$ with $g_2, g_3 \in \overline{\mathbb{Q}}$ we write $\wp, \zeta$ for its Weierstrass functions, Λ_E for its period lattice, $\eta(\lambda) = \zeta(z+\lambda) - \zeta(z)$ for the quasi-periods, and $\mathcal{L}_E = \{u \in \mathbb{C} : u \in \Lambda_E \text{ or } \wp(u) \in \overline{\mathbb{Q}}\}$. We use two curves throughout:

$$E_i\colon y^2 = 4x^3 - 4x, \qquad \Lambda = \varpi\,\mathbb{Z}[i], \qquad \varpi = \tfrac{\Gamma(1/4)^2}{2\sqrt{2\pi}}, \qquad \eta(\varpi) = \tfrac{\pi}{\varpi};$$
$$E_\rho\colon y^2 = 4x^3 - 4, \qquad \Lambda = \Omega_\rho\mathbb{Z}[\rho], \qquad \Omega_\rho = \tfrac{\Gamma(1/3)^3}{2^{4/3}\pi}, \qquad \eta(\Omega_\rho) = \tfrac{2\pi}{\sqrt{3}\,\Omega_\rho}.$$

(The quasi-periods follow from the Legendre relation and $\zeta(iz) = -i\zeta(z)$, $\zeta(\rho z) = \bar\rho\zeta(z)$. The real periods are Beta integrals, $\varpi = \int_1^\infty (x^3-x)^{-1/2}dx = \frac{1}{2}B(\frac{1}{4},\frac{1}{2})$ and $\Omega_\rho = \int_1^\infty (x^3-1)^{-1/2}dx = \frac{1}{3}B(\frac{1}{6},\frac{1}{2})$, and the Γ-forms follow from the reflection and duplication formulas; these are instances of the Chowla–Selberg formula [7].)

For a CM curve E let $E^{\natural}$ be its universal vector extension and $\Lambda(E^{\natural})$ the logarithms of its algebraic points, written in coordinates (z, w) (base, and conjugate-eigen fibre; §5). For E_i, E_ρ the fibre coordinate can be normalised so that $X = (u, \zeta(u)) \in \Lambda(E^{\natural})$ for every $u \in \mathcal{L}_E \setminus \Lambda_E$, and in general $w(X) - \zeta^*(z(X)) \in \overline{\mathbb{Q}}$ for $X \in \Lambda(E^{\natural})$ with $z(X) \notin \Lambda_E$, where $\zeta^* = \zeta - \kappa_1\,\mathrm{id}$ and $\eta^* = \eta - \kappa_1\,\mathrm{id}$ are the conjugate-eigen versions of ζ and η ($\kappa_1 \in \overline{\mathbb{Q}}$, $\kappa_1 = 0$ for E_i, E_ρ; §5).

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Four exponentials at CM periods. The four exponentials conjecture asserts that if $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and $t \notin \mathbb{Q}$, then $\alpha^t = e^{t\log\alpha}$ and $\beta^t = e^{t\log\beta}$ are not both algebraic. It is known for algebraic t (Gel'fond–Schneider) and in several conjugation-generic situations (Diaz [6]; Waldschmidt [30]). We know of no fixed transcendental t for which it was known for all pairs, including real pairs such as $(2, 3)$ [31].

Theorem 1. *Let $E/\overline{\mathbb{Q}}$ have complex multiplication, let $\Omega \neq 0$ be a period of E, $c \in \overline{\mathbb{Q}}^\times$ and $\alpha, \beta \in \overline{\mathbb{Q}}^\times$. Fix logarithms $\log\alpha$, $\log\beta$ and put $\alpha^z = e^{z\log\alpha}$, $\beta^z = e^{z\log\beta}$.*

(a) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^{c\Omega}$ and $\beta^{c\Omega}$ are not both algebraic.*
(b) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^{c\Omega/\pi}$ and $\beta^{c\Omega/\pi}$ are not both algebraic.*

For example (with the real logarithms), 2^{ϖ} and 3^{ϖ} are not both algebraic; nor are $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$, nor 2^{Ω_ρ} and 3^{Ω_ρ}. Since the four exponentials statement is symmetric under $t \leftrightarrow 1/t$ (if $\alpha^{1/t}$, $\beta^{1/t}$ are algebraic, apply the statement at t to them), the same holds at $t \in \overline{\mathbb{Q}}^\times/\Omega$ and $t \in \overline{\mathbb{Q}}^\times\pi/\Omega$. Theorem 1 consists of Corollaries 19 and 14 of Theorems 18 and 13 (§3). These are dimension bounds that also allow elliptic logarithms or fibre values in the second row. Another consequence is:

Corollary 2. *For every CM curve with period Ω, at least one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ is transcendental. On $y^2 = 4x^3 - 4x$: one of $e^{\pi^2} \approx 19333.69$, $e^{\pi\varpi} \approx 3779.84$, $e^{\pi^2/\varpi} \approx 43.12$ is transcendental.*

Part of Theorem 1 also follows from the six exponentials theorem and complex conjugation (Diaz [6]; Waldschmidt [30, Cor. 2.7]): for $t \in \mathbb{R} \cup i\mathbb{R}$ and pairs with $\log\alpha, \log\beta, \overline{\log\beta}$ linearly independent over $\mathbb{Q}$ (e.g. $(2, 1+2i)$), and, exchanging the roles, for pairs with $\log\alpha, \log\beta \in \mathbb{R}\cup i\mathbb{R}$ at any t with $1, t, \bar{t}$ linearly independent over $\mathbb{Q}$. Since every t in Theorem 1 is an algebraic multiple of a real number, these arguments together already cover every pair at $t \notin \mathbb{R} \cup i\mathbb{R}$ (see the overlap paragraph after Corollary 14). They do not apply to real or purely imaginary t — such as $t = \varpi$, ϖ/π, Ω_ρ — with pairs whose logarithms span a conjugation-stable space, such as $(2, 3)$, which is where our examples lie, nor to Corollary 2.

Two CM curves. The same mechanism, applied to two non-isogenous CM curves, gives a statement with no logarithm of $\mathbb{G}_{\mathrm{m}}$ at all (and a twisted variant, Theorem 29 in §5.4). Applied to periods, it also gives two statements of degree two about periods of the two curves (Corollary 31): $\pi, \varpi, \Omega_\rho, \varpi\Omega_\rho$ are linearly independent over $\overline{\mathbb{Q}}$, and so are $1, \varpi, \Omega_\rho, \varpi\Omega_\rho/\pi$. These are not consequences of the analytic subgroup theorem, and we flag them as the most delicate claims of the paper.

Theorem 3 (Theorem MX)**.** *Let $X \in \Lambda(E_i^\natural)$, $Y \in \Lambda(E_\rho^\natural)$, $a, t \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^\times$. If*

$$\big(z(X) + z(Y) + a\big) \cdot \beta\,\eta(\Omega_\rho) \;=\; \big(w(X) + w(Y) + t\big) \cdot \varpi,$$

then $a = 0$, $z(X) \in \mathbb{Q}(i)\varpi$ and $z(Y) \in \mathbb{Q}(\rho)\Omega_\rho$; in particular $\exp_{E_i}(z(X))$ and $\exp_{E_\rho}(z(Y))$ are torsion points. In particular, for $u \in \mathcal{L}_{E_i} \setminus \Lambda$ and $u' \in \mathcal{L}_{E_\rho} \setminus \Lambda_\rho$ not both torsion, $(u + u' + a)\,\beta\,\eta(\Omega_\rho) \neq (\zeta_i(u) + \zeta_\rho(u') + t)\,\varpi$. An analogous statement holds for any two non-isogenous CM elliptic curves (Remark 27).

Corollary 4. *For $u \in \mathcal{L}_{E_i} \setminus \mathbb{Q}(i)\varpi$ and $a, t \in \overline{\mathbb{Q}}$,*

$$\frac{\varpi\,\Omega_\rho\,(\zeta_i(u) + t)}{\pi\,(u + a)} \notin \overline{\mathbb{Q}}, \qquad \textit{equivalently} \qquad \frac{\zeta_i(u) + t}{u + a} \notin \overline{\mathbb{Q}} \cdot \frac{\pi^{5/2}}{\Gamma(1/4)^2\,\Gamma(1/3)^3}.$$

Symmetrically for non-torsion $u' \in \mathcal{L}_{E_\rho}$ with ζ_ρ.

For example, for the point with $\wp(u) = 2$ one has $u = 0.72694593546890819853\ldots$ and $\varpi\Omega_\rho\zeta(u)/(\pi u) = 3.76379062156897182773\ldots$ (computed independently twice, by theta functions and by the Laurent series of ζ; an illustration only). The case $X = Y = 0$ of Theorem 3 gives:

Corollary 5. $\Gamma(1/4)^2\,\Gamma(1/3)^3/\pi^{5/2}$ *is transcendental.*

This is immediate from Wüstholz's theorem (Remark 28), and probably also from the Wolfart–Wüstholz theorem on Beta values [35] (not checked); we found it stated nowhere, but do not claim it as new. Algebraic independence of $\pi, \Gamma(1/4), \Gamma(1/3)$ is open, so Corollary 4 does not follow from known algebraic independence results.

The single-curve version of the same count gives the following theorem, which turns out to be classical in substance.

Theorem 6 (Theorem M)**.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication by k, and let $\Omega \neq 0$ be a period. If $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ and $\ell \in \widetilde{\mathcal{L}}$ satisfy $i\pi\, u = \Omega\,\ell$, then $u = s\Omega$ and $\ell = s\,i\pi$ for some $s \in \overline{\mathbb{Q}}$; if $u \in \mathcal{L}_E$ then $s \in k$.*

Corollary 7. *On E_i (and analogously on every CM curve): (M1) $\wp(c\varpi/\pi) \notin \overline{\mathbb{Q}}$ for $c \in \overline{\mathbb{Q}}^\times$; (M2) $\wp(\varpi \log\alpha/\pi) \notin \overline{\mathbb{Q}}$ for $\alpha \in \overline{\mathbb{Q}}^\times$ not a root of unity; (M3) at least one of e^{π^2}, $\wp(\pi\varpi)$ is transcendental; more precisely, if $e^{\pi^2} \in \overline{\mathbb{Q}}$ then $\wp(c\pi\Omega) \notin \overline{\mathbb{Q}}$ for every $c \in \overline{\mathbb{Q}}^\times$ and every period Ω of every CM curve; (M3′) for $\alpha \in \overline{\mathbb{Q}} \setminus \{0,1\}$ and any branch, one of $e^{i\pi\log\alpha}$, $\wp(\varpi\log\alpha)$ is transcendental; (Mη) $\eta(\varpi)\,u \notin \widetilde{\mathcal{L}}$ for $u \in \mathcal{L}_{E_i} \setminus \mathbb{Q}(i)\varpi$.*

Novelty, honestly. With $\ell \in \mathcal{L}$, Theorem 6 is the CM case, with one row of Waldschmidt's matrix a period of $E \times \mathbb{G}_{\mathrm{m}}$, of his Conjecture 42 in [29] (four exponentials for $E \times \mathbb{G}_{\mathrm{m}}$). Its special case $\ell \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}} \cdot \pi i$, $u \in \mathcal{L}_E$, follows in one line from Laurent's [16, Cor. 3, p. 146]: for any CM lattice, a period $\omega \neq 0$ and $u \notin \mathbb{Q}\Lambda_E$ with $\wp(u) \in \overline{\mathbb{Q}}$, the numbers $1, \omega, \eta(\omega), 2\pi i, \omega H(u), 2\pi i u/\omega$ are $\overline{\mathbb{Q}}$-linearly independent, where $H = \zeta - \gamma z$ and $\gamma = 0$ when $g_2 g_3 = 0$. (M1) is one line from it ($2\pi i u/\Omega = 2ic$), and so is (Mη) modulo $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\varpi + \overline{\mathbb{Q}}\eta(\varpi) + \overline{\mathbb{Q}}\pi i$. The remaining items of Corollary 7 (which only use $u \in \mathcal{L}_E$) follow quickly from Laurent's identity and the analytic subgroup theorem (§4.2); we found none of them stated in print (Abe [1] claimed (M1); the preprint was withdrawn). Masser's theorem [10] ($1, \omega, \eta, 2\pi i, \log\sigma$ are $\overline{\mathbb{Q}}$-linearly independent on a CM curve, for $\sigma \in \overline{\mathbb{Q}}^\times$ not a root of unity) is linear and gives none of these; nor do the algebraic independence results of Brownawell–Kubota [12], which have six-exponentials strength, or the third-kind results surveyed in [34] (Wüstholz's linear independence of $1, \omega, \eta, \lambda(u_i, \omega)$). Theorem 1, Corollary 2, Theorem 3 and Corollary 4 we found nowhere; the four exponentials problem is listed as open, with no known case of this kind, in [23, p. 27] and [31]. For a recent unrefereed preprint, partly Lean-checked relative to stated classical inputs, on CM periods of a different kind (irrationality exponent 2 for real periods of CM curves over $\mathbb{Q}$, and $\mu(\varpi^n) = O(n\log 2n)$), see [4]. For Theorem 1 the closest published result is [28, Thm. 4.2] (§3). The nearest published statement we know is Waldschmidt's [28, Thm. 4.3]: a bilinear statement for two elliptic curves (non-isogenous, or equal with k-independent first column), but for 2×9 matrices of elliptic logarithms, without quasi-periods and without a 2×2 case. His "Further variations" [27] concerns 2×2 minors with entries in $\widetilde{\mathcal{L}}$ only (no elliptic functions). Laurent [16] never mixes two curves; Corollary 4 is the cross-curve analogue of his Cor. 3, with $\eta(\Omega_\rho)$ in place of $\eta(\varpi)$. Sources we could not consult are listed in §8.

Why MX is different. After multiplying by π, every relation in Theorem 6 with $u \in \mathcal{L}_E$ becomes a $\overline{\mathbb{Q}}$-linear relation among periods of 1-motives, because of Laurent's identity (1); this is why the analytic subgroup theorem proves it too. The hypothesis of Theorem 3 instead contains the

products $\Omega_\rho\,\zeta_i(u)$ and $\eta(\varpi)\,u'$, each pairing a period of one curve with a logarithm of the other. No CM identity linearises these; by the period conjecture they are genuine degree-two periods (§6). The extra strength comes from a second true degree-two period relation (the Legendre relation of the second curve), which the count sees as a second period on the line.

2. The subgroup theorem with period term, and the Roy–Waldschmidt count

Let G be a commutative algebraic group over $\overline{\mathbb{Q}}$ of positive dimension, $T_G(\mathbb{C})$ its tangent space with the $\overline{\mathbb{Q}}$-structure from G, $\Omega(G) = \ker \exp_G$ and $\Lambda(G) = \exp_G^{-1}(G(\overline{\mathbb{Q}}))$. For a number field $K \subset \mathbb{C}$, $\Omega_K(G), \Lambda_K(G)$ are their K-spans. Let $d_0(G)$ (resp. $d_1(G)$) be the largest d with a surjection $G \to \mathbb{G}_{\mathrm{a}}^d$ (resp. $\mathbb{G}_{\mathrm{m}}^d$) over $\overline{\mathbb{Q}}$, and $\alpha(G) = 2\dim G - 2d_0(G) - d_1(G)$. Fix a basis $\beta_1, \dots, \beta_n$ of $K/\mathbb{Q}$ and $p_\beta\colon T_{G^n} = T_G^n \to T_G$, $(z_j) \mapsto \sum \beta_j z_j$. Let H be the largest connected algebraic subgroup of G^n defined over $\overline{\mathbb{Q}}$ with $p_\beta(T_H) = 0$ and put $\alpha_K(G) = \alpha(G^n/H)/n$ [20, p. 360]. $S_K(G)$ is the set of subspaces $p_\beta(T_{G'})$, $G' \subset G^n$ an algebraic subgroup defined over $\overline{\mathbb{Q}}$; they are $\overline{\mathbb{Q}}$-rational.

Theorem 8 ([20, Théorème, p. 363])**.** *Let $\mathcal{E}$ be a finite-dimensional $\mathbb{C}$-space with a $\overline{\mathbb{Q}}$-structure, $\Psi\colon T_G(\mathbb{C}) \to \mathcal{E}$ a $\overline{\mathbb{Q}}$-rational linear map, $W \subset V \subset \mathcal{E}$ with W rational, and $\kappa = \dim_K(V \cap \Psi(\Omega_K(G)))$. If (i) V contains no nonzero $\Psi(S)$ with $S \in S_K(G)$, then*

$$\dim_K\big(V \cap \Psi(\Lambda_K(G))\big) \le (\alpha_K(G) - \kappa)\dim_{\mathbb{C}}(V/W).$$

By [20, Remarque 1], (i) holds when V has no nonzero $\overline{\mathbb{Q}}$-rational point (and then $W = 0$), since every $\Psi(S)$ is rational. We use only this case. Theorem 8 is used below only for the first proof of Theorem 6 and in the sketch of §5.6. Theorems 13, 18 and 3 are proved from the following theorem directly; the sketch of Theorem 29 and the second derivations of Remarks 15, 20 and 26 use its hyperplane consequence, Theorem 11 below.

Theorem 9 ([21, Thm. 4.1, pp. 385–386])**.** *Let $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d = d_0 + d_1 + d_2 \ge 1$, with projections π_0, π_1 onto the first two factors (d_0, d_1 need not be maximal). Let $V \subset T_G(\mathbb{C})$ be a subspace of dimension $N < d$, $W \subset V$ a subspace defined over $\overline{\mathbb{Q}}$, and $Y \subset V$ a finitely generated subgroup with $\Gamma = \exp_G Y \subset G(\overline{\mathbb{Q}})$; put $\kappa = \mathrm{rk}_{\mathbb{Z}}(Y \cap \ker \exp_G)$. Then there is a connected algebraic subgroup $G' \ne G$ defined over $\overline{\mathbb{Q}}$ such that, with $\delta = \dim G/G'$, $\delta_0 = \dim \mathbb{G}_{\mathrm{a}}^{d_0}/\pi_0(G')$, $\delta_1 = \dim \mathbb{G}_{\mathrm{m}}^{d_1}/\pi_1(G')$, $\delta_2 = \delta - \delta_0 - \delta_1$, $\lambda = \mathrm{rk}_{\mathbb{Z}}\,\Gamma/(\Gamma \cap G')$ and $\tau = \dim W/(W \cap T_{G'})$, one has $\delta > \tau$ and*

$$(\lambda + \delta_1 + 2\delta_2)(d - N) \le (\delta - \tau)(d_1 + 2d_2 - \kappa).$$

($G' = 0$ is allowed.) The proof [21, §§6–7] combines an auxiliary function [21, Prop. 6.1] with Philippon's zero estimate [11]. Prop. 6.1 asks for a polynomial that vanishes with multiplicity $\ge T$ along W at the points of $\exp_G(Y(S))$, with the saving κ in the parameters. The κ-saving is proved in [26, Prop. 2.1 and 2.4] (an auxiliary function constructed with periods), for $W = 0$ and $b = 1$ [21, p. 396]; for the rest W88 refers to [26] and to Bertrand [25, App. I, Lemme 7 and its Corollaire, with Lemme 1] (cited in [21, p. 396] as "Lemma 7"). All our applications use both $W \ne 0$ and $\kappa > 0$. We write out the multiplicity step in Appendix A (Lemma 32). It holds under the condition $2N \ge d_1 + 2d_0 + \kappa$, which is satisfied in all the applications in §§3 and 5 (for Theorem 13: $5n - 2 \ge 3n$; Theorem 18: $5n \ge 3n$; Theorem 3: $4n + 2 \ge 2n + 2$; Theorem 29: $4n + 2 \ge 2n$; Theorem 21: $5n + 2 \ge 3n + 4$).

We use one piece of linear algebra repeatedly.

Lemma 10. *Let $k \subset K$ be an imaginary quadratic field inside a number field of degree n, $\theta_1, \dots, \theta_n$ a basis of $K/\mathbb{Q}$, and on k^n put $\psi(m) = \sum \theta_j m_j$, $\phi(m) = \sum \theta_j \bar{m}_j$. Then $(\psi, \phi)\colon k^n \to$*

$K \times K$ is a $\mathbb{Q}$-linear bijection; $\ker\psi$ *and* $\ker\phi$ *are k-subspaces of dimension $n/2$, and* $\ker\psi \cap \ker\phi = 0$.

Proof. $k^n \cong K \otimes_{\mathbb{Q}} k$ via $m \mapsto \sum \theta_j \otimes m_j$, and $K \otimes_{\mathbb{Q}} k \cong K \times K$ via $x \otimes y \mapsto (xy, x\bar{y})$; the composite is (ψ, ϕ). ψ is k-linear and ϕ is k-semilinear, and both are onto K, which has k-dimension $n/2$. □

Where Theorem 8 is proved, and the dependence. The proof [20, pp. 363–364] has two steps.

(b) (Reduction, fully written.) With $\Psi_\beta = \Psi \circ p_\beta$, let H be the largest connected subgroup with $T_H \subset \ker\Psi_\beta$, $G' = G^n/H$, $\Psi'\colon T_{G'} \to \mathcal{E}$ the induced map, $V' = \Psi'^{-1}(V)$, $W' = \Psi'^{-1}(W)$. Then $\dim_{\mathbb{Q}}(V' \cap \Lambda(G')) = n \dim_K(V \cap \Psi(\Lambda_K))$ and $\dim_{\mathbb{Q}}(V' \cap \Omega(G')) = n\kappa$, and the theorem follows from case (a) for G' and $\alpha(G') \le n\alpha_K(G)$.

(a) (Case $K = \mathbb{Q}$, $\Psi = \mathrm{id}$.) *Statement R*: if $V \subset T_G(\mathbb{C})$ contains no nonzero Lie algebra of an algebraic subgroup of G defined over $\overline{\mathbb{Q}}$, and $W \subset V$ is $\overline{\mathbb{Q}}$-rational, then

$$\dim_{\mathbb{Q}}(V \cap \Lambda(G)) \le \big(\alpha(G) - \dim_{\mathbb{Q}}(V \cap \Omega(G))\big) \dim_{\mathbb{C}}(V/W).$$

RW93 derive (i)⇒(iii) here "de manière purement formelle" from Theorem 4.1 of Waldschmidt [21] (the algebraic subgroup theorem with the period rank $\kappa = \mathrm{rk}(Y \cap \ker\exp_G)$; we checked its statement on pp. 385–386), by applying Roy's categorical Proposition 2.1 of [18] to the category $\mathcal{G}_w$ and the functions a', b, c, d, r of [18, §6]. Their Remark 2 says that Statement R with the κ-term was known when $\dim V = \dim G - 1$ (citing [21, Thm. 1.1]) or with κ replaced by 0 ([18, §6, Remark (i)]), and is proved in general "en raffinant les arguments du §6 de [R1] (en travaillant avec la fonction a' au lieu de a'')".

Step (b) also uses case (a) for $\ker\Psi'$, to see that Ψ' is injective on $\Lambda(G')$; this also follows from Wüstholz's analytic subgroup theorem [36].

The dependence, stated precisely. When $V' = \Psi'^{-1}(V)$ is a **hyperplane** of $T_{G'}$ (as in all our applications), Statement R with the κ-term is in substance the following refinement of [21, Thm. 1.1], proved in [21, §5d, p. 394].

Theorem 11 ([21, Thm. 1.1, pp. 375–376, under (5.4) of §5d, p. 394])**.** *Let $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d \ge 1$, with d_0, d_1 maximal and $d_2 = d - d_0 - d_1$. Let V be a hyperplane of $T_G(\mathbb{C})$, $W \subset V$ a subspace of dimension $t \ge 0$ defined over $\overline{\mathbb{Q}}$, and $Y = \mathbb{Z}y_1 + \cdots + \mathbb{Z}y_m \subset V$ a subgroup of rank m with $\exp_G Y \subset G(\overline{\mathbb{Q}})$. Put $\kappa = \mathrm{rk}_{\mathbb{Z}}(Y \cap \ker\exp_G)$. If*

$$m > (d_1 + 2d_2 - \kappa)(d - 1 - t), \tag{5.4}$$

then V contains a nonzero algebraic Lie subalgebra of $T_G(\mathbb{C})$ defined over $\overline{\mathbb{Q}}$, that is, $T_{G'} \subset V$ for some connected algebraic subgroup $G' \ne 0$ of G defined over $\overline{\mathbb{Q}}$.

The printed Theorem 1.1 has the hypothesis $m > (d_1 + 2d_2)(d - 1 - t)$, without κ; [21, §5d] proves it under the weaker hypothesis (5.4) (p. 394 prints $\kappa = \ell - m$, a misprint for $m - \ell$, cf. Lemma 5.2 there). The case $W = 0$ ($t = 0$) of Theorem 11 is also printed as [28, Thm. 5.1, p. 352], where it is attributed to [21, Thm. 1.1] and to [32, Thm. 4.1]. The proof in [21, §5d] applies Theorem 9 (in the form [21, Cor. 5.3]) to G with $N = d - 1$, and then again to an auxiliary subgroup G' given by [21, Lemma 5.1]. The condition of Lemma 32 below covers the first application in the cases listed above; we have not checked it for the second. So every use of Theorem 11 rests on [21, §5d] as published.

For Theorems 13, 18 and 3 the main proofs bypass Statement R: §§3, 5 apply Theorem 9 to explicit quotient groups and exclude every connected algebraic subgroup. Each of the three also

has a second derivation from Theorem 11, with a two-line subgroup exclusion (Remarks 15, 20 and 26). The sketch of Theorem 29 uses Theorem 11 (a full case check against Theorem 9 is only mentioned there). For Theorem 6, the cases (M1), (M3) were checked against Theorem 9 in the same way, and the second proof (§4.2) avoids [20] for $u \in \mathcal{L}_E$; only $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ uses Theorem 8 as stated.

Weights used. $\alpha_K(\mathbb{G}_a) = 0$, $\alpha_K(\mathbb{G}_m) = 1$, and for an elliptic curve $\alpha_K(E) = 1$ if E has CM by k and $K \supseteq k$, else 2 [20, Lemme, p. 361]; α_K is additive on products [20, (2)]. For the universal vector extension $E^\natural$ of a CM curve and $K \supseteq k$ we need:

Lemma 12. *$\alpha_K(E^\natural) = 2 + 2/n$.*

Proof. Use coordinates on $T(E^\natural) = H^1_{\mathrm{dR}}(E)^\vee$ in which $\mathcal{O} = \mathrm{End}(E)$ acts by its identity character on the base coordinate and by the conjugate character on the fibre coordinate (the eigenline of H^1_{dR} for $\bar a$ is spanned by $x\,dx/y + c\,dx/y$ with $c \in \overline{\mathbb{Q}}$, so these coordinates are rational). Let $H \subset (E^\natural)^n$ be connected with $p_\beta(T_H) = 0$, with abelian image $B \subset E^n$. Then $B = B_U$ for a k-subspace $U \subset k^n$ on which both $u \mapsto \sum \beta_j u_j$ (base) and $u \mapsto \sum \beta_j \bar u_j$ (fibre) vanish. The map $k^n \to K \times K$, $u \mapsto (\sum \beta_j u_j, \sum \beta_j \bar u_j)$ is injective (it is the composite $k^n \cong K \otimes_{\mathbb{Q}} k \cong K \times K$, $x \otimes y \mapsto (xy, x\bar y)$), so $B = 0$. Hence H lies in the fibre $\mathbb{G}_a^n$, and the largest such H is the rational hyperplane $\sum \beta_j t_j = 0$, of dimension $n-1$. The quotient has dimension $n+1$ and admits no surjection to $\mathbb{G}_a$ or $\mathbb{G}_m$ (it is a non-split extension of E^n by $\mathbb{G}_a$), so $\alpha = 2(n+1)$ and $\alpha_K = 2 + 2/n$. □

3. Four exponentials at CM periods

Throughout this section $E/\overline{\mathbb{Q}}$ has complex multiplication by k, $\Omega \neq 0$ is a period of E, and for a number field K we write $K\mathcal{L}$, $K\mathcal{L}_E$ for the K-spans of $\mathcal{L}$ and $\mathcal{L}_E$. By Baker's theorem $K\mathcal{L} \cong K \otimes_{\mathbb{Q}} \mathcal{L}$, so elements of $\mathcal{L}$ that are $\mathbb{Q}$-linearly independent are K-linearly independent.

Theorem 13 (Theorem MG)**.** *Let $c \in \overline{\mathbb{Q}}^\times$ and $s = c\,\Omega/\pi$. For every number field $K \supseteq k$,*

$$\mathcal{S}_K := \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K\mathcal{L}_E\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K \le 2,$$

and $2\pi i \in \mathcal{S}_K$ (since $s \cdot 2\pi i = 2ic\,\Omega \in K\mathcal{L}_E$ when $ic \in K$).

Proof. Enlarging K does not change the conclusion (by Baker, K-independence in $K\mathcal{L}$ is preserved under base change), so assume $K \supseteq k(i, c)$; $n = [K : \mathbb{Q}]$ is even (the count below closes for every such n). Let $G = \mathbb{G}_m \times \mathbb{G}_m \times E$ with coordinates (x, y, z), $\Psi(x, y, z) = (x, y + z)$, $V = \mathbb{C}(1, s)$. Fix a basis $\theta_1, \dots, \theta_n$ of $K/\mathbb{Q}$, put $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, and $\Phi = \Psi \circ p_\theta$.

The group. Connected algebraic subgroups of $\mathbb{G}_m^{2n} \times E^n$ are products $T_L \times B_S$ of a subtorus (given by a subspace $L \subset \mathbb{Q}^{2n}$) and an abelian subvariety (given by a k-subspace $S \subset k^n$). Such a subgroup has tangent space in $\ker \Phi$ iff $L = 0$ and $\psi(S) = 0$, where $\psi(m) = \sum \theta_j m_j$; by Lemma 10, $\ker \psi$ has k-dimension $n/2$. Put $U = B_{\ker \psi}$ and $G'' = G^n/U = \mathbb{G}_m^{2n} \times A''$, $A'' = E^n/U$, of dimension $n/2$. Then $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2$. The induced map Ψ'' is onto $\mathbb{C}^2$, $V'' = \Psi''^{-1}(V)$ is a hyperplane, and $W'' = \ker \Psi'' \subset V''$ is $\overline{\mathbb{Q}}$-rational of dimension $d - 2$. For a connected $G' \subset G''$ put $r' = \dim \Psi''(T_{G'})$; then $\tau = \delta + r' - 2$, so only $r' \le 1$ can satisfy $\delta > \tau$, and $\delta - \tau = 2 - r'$.

The data. $w = (2\pi i, 2ic\Omega) \in \Psi(\Omega_K(G))$. Suppose $x_1, x_2 \in \mathcal{S}_K$ with $2\pi i, x_1, x_2$ K-independent, and put $v_j = (x_j, s x_j) \in V \cap \Psi(\Lambda_K(G))$. Let Y'' be generated by the images in $T_{G''}$ of elements of $\Lambda(G^n)$ with Φ-images $D\theta_h w$ (taken in $\Omega(G^n)$, D a common denominator), $\theta_h v_1$, $\theta_h v_2$ ($h = 1, \dots, n$). Then $\mathrm{rk}\, Y'' = 3n$, $\Gamma = \exp Y'' \subset G''(\overline{\mathbb{Q}})$, and $\kappa = n$: a period in Y'' has first coordinate in $2\pi i K$, so its v-parts vanish. The inequality of Theorem 9 reads $\lambda + \delta_1 + 2\delta_2 \le (2 - r') \cdot 2n$.

No obstruction.

- $G' = 0$: $\lambda = 2n$, so the left side is $2n + 2n + n = 5n > 4n$.
- $r' = 1$ with image $\mathbb{C}(0,1)$ (this covers every G' with an abelian part, since ψ does not vanish on any nonzero abelian subvariety of A''): an element of Y'' lying in $T_{G'} + \Omega(G'')$ has first coordinate in $2\pi i K$, so $\mathrm{rk}(Y'' \cap (T_{G'} + \Omega(G''))) \le n$ and $\lambda \ge 2n$; the left side is $\ge 3n > 2n$.
- $r' = 1$ with image a line $\mathbb{C}(p,q) \ne \mathbb{C}(0,1)$: then $G' = T_L$ with $\dim L \le n$. Two K-independent data in $T_{G'} + \Omega(G'')$ would put x_1, x_2 in a K-space of dimension ≤ 2 containing $2\pi i$ (namely $K \cdot 2\pi i + K\pi^2/\Omega$ if $q = 0$, and $K \cdot 2\pi i + K \cdot p\,\Omega/(sp-1)$ otherwise, using $s \notin K$), against the K-independence of $2\pi i, x_1, x_2$. So $\lambda \ge n$, $\delta_1 \ge n$, $\delta_2 = n/2$, and the left side is $\ge 3n > 2n$.
- $r' = 0$ forces $G' = 0$.

So no connected $G' \ne G''$ satisfies the conclusion of Theorem 9, a contradiction. □

Controls. If E has no CM then $U = 0$, $d_2 = n$, and $G' = 0$ satisfies the inequality ($6n \le 6n$); so does $G' = 0$ with κ deleted ($5n \le 6n$), with only one datum, or with $k \not\subset K$. If $s \in K$ (a rational line) the full subtorus over V satisfies it with equality; replacing E by $\mathbb{G}_{\mathrm{m}}$ makes s algebraic, and then the statement is false ($s = 2$, $x = \log 2$).

Corollary 14 (MG2: four exponentials at $t \in \overline{\mathbb{Q}}^{\times}\Omega/\pi$)**.** *Let $t \in \overline{\mathbb{Q}}^{\times} \cdot \Omega/\pi$ and $\alpha, \beta \in \overline{\mathbb{Q}}^{\times}$ with $\log\alpha, \log\beta$ linearly independent over $\mathbb{Q}$. Then $\alpha^t = e^{t\log\alpha}$ and $\beta^t = e^{t\log\beta}$ are not both algebraic. If moreover $\log\alpha, \log\beta, 2\pi i$ are linearly independent over $\mathbb{Q}$, then $t\log\alpha$ and $t\log\beta$ are not both in $\overline{\mathbb{Q}}\mathcal{L} + \overline{\mathbb{Q}}\mathcal{L}_E$. For example, $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$ are not both algebraic.*

Proof. If $\log\alpha, \log\beta, 2\pi i$ are $\mathbb{Q}$-independent, take $x_1 = \log\alpha$, $x_2 = \log\beta$ in Theorem 13: with $2\pi i$ they are three $\mathbb{Q}$-independent logarithms, hence K-independent. Otherwise $m\log\beta = a\log\alpha + 2\pi i\, b$ with integers $m \ne 0$, $b \ne 0$. If α^t and β^t are algebraic, so is $e^{2\pi i b t} = (\beta^t)^m(\alpha^t)^{-a}$, i.e. $2\pi i b t = 2ib\,c\,\Omega \in \mathcal{L}$ (with $t = c\Omega/\pi$). Then $\Omega \in \overline{\mathbb{Q}}\mathcal{L}$, which the analytic subgroup theorem [36] excludes (on $\mathbb{G}_{\mathrm{m}} \times E$, the smallest algebraic subgroup whose tangent space contains (ℓ, Ω) is $\mathbb{G}_{\mathrm{m}} \times E$). (Remark 15 gives a proof that needs no such step.) □

The determinations matter: $\log\alpha = \log 2$ and $\log\beta = \log 2 + 2\pi i$ are $\mathbb{Q}$-independent, and for the determinations $e^{t\log 2}$ of both powers the statement would say $2^t \notin \overline{\mathbb{Q}}$, a one-exponential statement that we do not prove.

Remark 15 (Second derivation via §5d)**.** Theorem 11 gives Theorem 13 with a shorter subgroup analysis, and the first sentence of Corollary 14 without the case distinction. Keep $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, Ψ'', $V'' = \Psi''^{-1}(V)$ and $W'' = \ker\Psi''$ from the proof of Theorem 13: $d = 5n/2$, $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, and W'' is $\overline{\mathbb{Q}}$-rational of dimension $t = d - 2$, so $d - 1 - t = 1$. Let Y'' be generated by the $3n$ lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$ ($h = 1, \dots, n$). The n period lifts lie in $Y'' \cap \ker\exp$, so $\kappa \ge n$ (in fact $\kappa = n$), and (5.4) reads $3n > (2n + n - \kappa) \cdot 1$, i.e. $3n > 2n$.

The rank $m = 3n$. For the data of Theorem 13 ($2\pi i, x_1, x_2$ K-independent) the first coordinates of the Ψ''-images $D\theta_h \cdot 2\pi i$, $\theta_h x_j$ are $\mathbb{Q}$-independent. For the data of Corollary 14 ($x_j = \log\alpha_j$, $sx_j = t\log\alpha_j \in \mathcal{L}$) only the $\mathbb{Q}$-independence of $\log\alpha, \log\beta$ is needed. Take the period lifts with toric part $(2\pi i D e_h, 0)$ and E^n-part $m^{(h)}\Omega \in \Lambda_E^n$, where $\psi(m^{(h)}) = 2icD\theta_h$ (possible by Lemma 10, as $ic \in K$; D a common denominator), and the data lifts toric, $(x_j e_h,\ sx_j e_h)$. The linear form induced by $\theta^{\mathsf{T}} z$ on $T_{A''}$ takes the $\mathbb{Q}$-independent values $2icD\theta_h\Omega$ on the period lifts and 0 on the data lifts, so a $\mathbb{Q}$-relation among the $3n$ lifts involves only the data lifts; their first-block coordinates $x_j e_h$ are $\mathbb{Q}$-independent.

No subgroup in V''. Theorem 11 gives a connected $G' \ne 0$ with $T_{G'} \subset V''$, and $G' = T_L \times B_S$. If $B_S \ne 0$, then $T_{G'}$ contains a vector $(0, b)$ with $\theta^{\mathsf{T}} b \ne 0$ (since ψ does not vanish on any nonzero abelian subvariety of A''), whose image $(0, \theta^{\mathsf{T}} b)$ is not on $V = \mathbb{C}(1, s)$. If $B_S = 0$, then $L \ne 0$,

and for $0 \neq l = (l_1, l_2) \in L$ the image $(\theta^{\mathsf{T}} l_1, \theta^{\mathsf{T}} l_2) \in K^2 \setminus 0$ lies on $\mathbb{C}(1, s)$; so $\theta^{\mathsf{T}} l_1 \neq 0$ and $s \in K$, which is false ($\Omega/\pi \notin \overline{\mathbb{Q}}$ [17]).

For $K = \mathbb{Q}(i)$, $n = 2$, $\theta = (1, i)$: $A'' \cong E$ via $(z_1, z_2) \mapsto z_1 + iz_2$, so $G'' = \mathbb{G}_{\mathrm{m}}^4 \times E$, $d = 5$, $t = 3$, $m = 6$, $\kappa = 2$, and (5.4) is $6 > 6 - \kappa$. This route uses neither the K-independence of $2\pi i, \log\alpha, \log\beta$ nor Wüstholz's theorem: it proves the first sentence of Corollary 14 directly, also when $2\pi i$ lies in the $\mathbb{Q}$-span of $\log\alpha, \log\beta$. Its subgroup exclusion uses only $s \notin K$. It rests on Theorem 11 as published, where the main proof rests on Theorem 9 and Lemma 32.

Corollary 16 (MG1)**.** *For every CM curve E, every period Ω and all $a, b, c \in \overline{\mathbb{Q}}^{\times}$, at least one of $e^{a\pi^2}$, $e^{b\pi\Omega}$, $e^{c\pi^2/\Omega}$ is transcendental. On E_i: one of e^{π^2}, $e^{\pi\varpi}$, $e^{\pi\eta(\varpi)}$ is transcendental.*

Proof. Take $s = \Omega/\pi$ (up to $\overline{\mathbb{Q}}^{\times}$), $x_1 = \pi^2$, $x_2 = \pi^2/\Omega$: if all three exponentials are algebraic then $x_1, x_2 \in \overline{\mathbb{Q}}\mathcal{L}$ and $sx_1 = \pi\Omega$, $sx_2 = \pi$ lie in $\overline{\mathbb{Q}}\mathcal{L}$. The numbers $2\pi i, \pi^2, \pi^2/\Omega$ are $\overline{\mathbb{Q}}$-independent. A relation without the π^2/Ω term contradicts the transcendence of π. Otherwise $\pi/\Omega = a_1 + b_1\pi$ with $a_1, b_1 \in \overline{\mathbb{Q}}$. If $b_1 = 0$ then $\Omega/\pi \in \overline{\mathbb{Q}}$, which is false. If $b_1 \neq 0$ then $\pi\Omega = (\pi - a_1\Omega)/b_1$; since $\pi\Omega \in \overline{\mathbb{Q}}\mathcal{L}$ (as $e^{b\pi\Omega} \in \overline{\mathbb{Q}}$) and $\pi \in \overline{\mathbb{Q}}\mathcal{L}$, this gives $a_1\Omega \in \overline{\mathbb{Q}}\mathcal{L}$, so $a_1 = 0$ because $\Omega \notin \overline{\mathbb{Q}}\mathcal{L}$ (proof of Corollary 14); then $\Omega = 1/b_1 \in \overline{\mathbb{Q}}$, which is false. (Chudnovsky's algebraic independence of π and Ω [5] would give this at once, but is not needed.) On E_i, $\pi/\varpi = \eta(\varpi)$. □

For a general CM curve, $\eta(\Omega) = \kappa_1\Omega + \kappa_2\pi/\Omega$ with $\kappa_1, \kappa_2 \in \overline{\mathbb{Q}}$ and $\kappa_2 \neq 0$ (Legendre); $\kappa_1 = 0$ when $j(E) \in \{0, 1728\}$. Since $\pi\Omega, \pi\eta \in \overline{\mathbb{Q}}\mathcal{L}$ imply $\pi^2/\Omega = (\pi\eta - \kappa_1\pi\Omega)/\kappa_2 \in \overline{\mathbb{Q}}\mathcal{L}$, the statement "one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi\eta}$ is transcendental" holds for every CM period as well.

Corollary 17 (MG3)**.** *If e^{π^2} and $e^{c\pi\Omega}$ ($c \in \overline{\mathbb{Q}}^{\times}$) are algebraic, then $\alpha^{\Omega/\pi} \notin \overline{\mathbb{Q}}$, for every determination, for every $\alpha \in \overline{\mathbb{Q}}^{\times}$ such that $2\pi i, \pi^2, \log\alpha$ are linearly independent over $\mathbb{Q}$.*

Proof. With $s = \Omega/\pi$, take $x_1 = \pi^2$ (then $sx_1 = \pi\Omega \in \overline{\mathbb{Q}}\mathcal{L}$) and $x_2 = \log\alpha$ (then $sx_2 \in \mathcal{L}$ if $\alpha^{\Omega/\pi} \in \overline{\mathbb{Q}}$). These are $\mathbb{Q}$-independent logarithms together with $2\pi i$, hence K-independent, and Theorem 13 gives a contradiction. The hypothesis is needed: for $c = 1$ and $\alpha = e^{\pi^2}$ ($\log\alpha = \pi^2$) one has $\alpha^{\Omega/\pi} = e^{\pi\Omega} \in \overline{\mathbb{Q}}$. (Corollary 17 is also the case $\beta = e^{\pi^2}$ of the second sentence of Corollary 14.) □

Overlap with known results. Waldschmidt's [30, Cor. 2.7] (after Diaz [6]) says: if $x_1, x_2 \in \mathbb{R} \cup i\mathbb{R}$ are linearly independent over $\mathbb{Q}$ and $y_1, y_2, \bar{y}_2$ are linearly independent over $\mathbb{Q}$, then one of the four numbers $e^{x_i y_j}$ is transcendental. With $x = (1, t)$ it gives Corollaries 14 and 19 for $t \in \mathbb{R} \cup i\mathbb{R}$ and pairs with $\log\alpha, \log\beta, \overline{\log\beta}$ $\mathbb{Q}$-independent (e.g. $(2, 1 + 2i)$). With the roles exchanged, $x = (\log\alpha, \log\beta)$ and $y = (1, t, \bar{t})$, it gives them for pairs with logarithms in $\mathbb{R} \cup i\mathbb{R}$ (e.g. $(2, 3)$) whenever $1, t, \bar{t}$ are $\mathbb{Q}$-independent, e.g. $t = e^{i\pi/5}\varpi/\pi$. Rescaling widens the first orientation to every t in our classes. Such a t is γr with $\gamma \in \overline{\mathbb{Q}}^{\times}$ and $r \in \mathbb{R}$ ($\bar{E}$ has CM by the same field, so $\bar{\Omega} \in \overline{\mathbb{Q}}^{\times}\Omega$; and π is real). If $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$, then $y = (\log\alpha, \log\beta)$ satisfies $y_j,\ ry_j = \gamma^{-1} t y_j \in \widetilde{\mathcal{L}}$, and the $\widetilde{\mathcal{L}}$-version [30, Thm. 4.1] at the real point r excludes every pair with $\log\alpha, \log\beta, \overline{\log\beta}$ linearly independent (over $\overline{\mathbb{Q}}$, which for logarithms is the same as over $\mathbb{Q}$ by Baker's theorem), whatever the argument of t. The remaining pairs have a conjugation-stable $\mathbb{Q}$-span of logarithms, which has a basis in $\mathbb{R} \cup i\mathbb{R}$. For them the second orientation applies as soon as $t \notin \mathbb{R} \cup i\mathbb{R}$: then $1, t, \bar{t}$ are $\mathbb{Q}$-independent, because r is transcendental and $\bar{\gamma}/\gamma \notin \mathbb{Q}$. So at $t \notin \mathbb{R} \cup i\mathbb{R}$ Corollaries 14 and 19 are already known. What remains new is real or purely imaginary t (as for ϖ, ϖ/π, Ω_ρ) with a conjugation-stable span of logarithms (such as $(2, 3)$, or $(5, (1 + 2i)/(1 - 2i))$); Corollary 16; and the dimension statements themselves. We found no case of the four exponentials conjecture at a fixed transcendental t of this kind in [31] or [23, p. 27]. The closest published theorem is Waldschmidt's [28, Thm. 4.2], a mixed statement on $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times E$ without CM whose second

row consists of elliptic logarithms; it needs seven columns (one of six numbers $\ell_1 u_j - \ell_j u_1$ is transcendental) in general, and five columns (one of four) in its period-column case. CM and the mixing map bring the count down to two data with ordinary logarithms.

3.1. **The fibre twin.** Let $E^\natural$ be read on its conjugate-eigen fibre coordinate w (§5), with $\eta^* = \eta^*(\Omega)$. By Legendre and $\eta^*(a\lambda) = \bar{a}\eta^*(\lambda)$, $\Omega\eta^* \in \overline{\mathbb{Q}}^\times\pi$ for every CM curve. Call *fibre values* the w-coordinates of $\Lambda(E^\natural)$ (that is, $\zeta^*(u) + \overline{\mathbb{Q}}$ for $u \in \mathcal{L}_E \setminus \Lambda_E$, $\eta^*(\lambda) + \overline{\mathbb{Q}}$ at lattice points).

Theorem 18 (Theorem MG$^\natural$)**.** *Let $c \in \overline{\mathbb{Q}}^\times$ and $s = c\,\eta^*/\pi \in \overline{\mathbb{Q}}^\times/\Omega$. For every number field $K \supseteq k$, $\dim_K\{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K \cdot (\textit{fibre values})\} \le 2$.*

Proof. As in the proof of Theorem 13 we may enlarge K; we take $K \supseteq k(i, c)$ with $n = [K : \mathbb{Q}] \ge 4$ (for $n = 2$ the count below does not close). Let $G = \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m} \times E^\natural$, $\Psi(x, y, X) = (x,\, y + w(X))$, $V = \mathbb{C}(1, s)$, and θ, p_θ, $\Phi = \Psi \circ p_\theta$ as before.

The group. There are no nonzero homomorphisms between tori and connected subgroups of $(E^\natural)^n$ in either direction, so connected subgroups of $\mathbb{G}_\mathrm{m}^{2n} \times (E^\natural)^n$ are products. A connected $H \subset (E^\natural)^n$ over B_S contains the canonical lift $B_S^\natural$ (since $\mathrm{Hom}(B_S^\natural, \mathbb{G}_\mathrm{a}) = 0$), whose tangent space is $\{(\sigma(M)z, \bar{\sigma}(M)w)\}$; so $\theta^\mathsf{T} w$ vanishes on T_H only if $\phi(S) = 0$. Hence the largest connected U with $T_U \subset \ker\Phi$ is $U = B_{\ker\phi}^\natural + \{\theta^\mathsf{T} w = 0\}$, and $G'' = \mathbb{G}_\mathrm{m}^{2n} \times Q''$ with $Q'' = (E^\natural)^n/U$, an extension of $A'' = E^n/B_{\ker\phi}$ (dimension $n/2$) by $\mathbb{G}_\mathrm{a}$ (coordinate $\sigma = \theta^\mathsf{T} w$). Over an abelian subvariety $B' \ne 0$ of A'', coming from $S' \supsetneq \ker\phi$, we have $\phi(S') \ne 0$, so the image of $B_{S'}^\natural$ contains the fibre. Thus Q'' has no splitting over any $B' \ne 0$, and its connected subgroups are 0, the fibre F, and the $\pi^{-1}(B')$ (dimension $\le n/2 + 1$). Q'' is a quotient of an anti-affine group, so $\mathrm{Hom}(Q'', \mathbb{G}_\mathrm{a}) = 0$: $d_0 = 0$, $d_1 = 2n$, $G_2 = Q''$, $d_2 = n/2 + 1$. The induced map is $\Psi''(t_1, t_2, q) = (\theta^\mathsf{T} t_1, \theta^\mathsf{T} t_2 + \sigma(q))$, V'' is a hyperplane, $W'' = \ker\Psi''$, and $\tau = \delta + r' - 2$ as before.

The data. The period datum is $(2\pi i, 2ic\eta^*) \in \Psi(\Omega_K(G))$ (using $ic \in K$ and that ϕ is onto K). With x_1, x_2 as in Theorem 13, $\mathrm{rk}\,Y'' = 3n$ and $\kappa = n$ (a period of G'' has first coordinate in $2\pi iK$). The inequality reads $\lambda + \delta_1 + 2\delta_2 \le (2 - r')(2n + 2)$.

No obstruction. Every G' is $T_L \times H$ with $H \in \{0, F, \pi^{-1}(B')\}$, and $H \ne 0$ forces $\mathbb{C}(0, 1) \subset \Psi''(T_{G'})$.

- $G' = 0$: $\lambda = 2n$, and the left side is $2n + 2n + (n + 2) = 5n + 2 > 4n + 4$.
- Image $\mathbb{C}(0, 1)$ ($H \ne 0$, or L in the second block; $l = \dim L \le n$, $h = \dim H \le n/2 + 1$): elements of Y'' in $T_{G'} + \Omega(G'')$ have first coordinate in $2\pi iK$, so $\lambda \ge 2n$, and the left side is $\ge 2n + (2n - l) + 2(n/2 + 1 - h) \ge 3n > 2n + 2$.
- $H = 0$ and image a line $\mathbb{C}(1, q)$, $q \in K$: a datum X in $T_{G'} + \Omega(G'')$ satisfies $X(s - q) \in K{\cdot}2\pi i + K{\cdot}2\pi is$, so $X \in K{\cdot}2\pi i + K{\cdot}2\pi i/(s - q)$ (here $s \notin K$ because Ω is transcendental). So $\lambda \ge n$, and the left side is $\ge n + n + (n + 2) = 3n + 2 > 2n + 2$.
- $r' = 0$ forces $G' = 0$; $r' = 2$ fails $\delta > \tau$.

The minimum margin is $n - 2$. □

Corollary 19 (MG$^\natural$2: four exponentials at $t \in \overline{\mathbb{Q}}^\times\Omega$)**.** *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and $c \in \overline{\mathbb{Q}}^\times$, then $\alpha^{c\Omega} = e^{c\Omega\log\alpha}$ and $\beta^{c\Omega} = e^{c\Omega\log\beta}$ are not both algebraic. For example, 2^ϖ and 3^ϖ (real logarithms) are not both algebraic.*

Proof. Put $x_j = c\Omega\log\alpha_j$ (with $\alpha_1 = \alpha$, $\alpha_2 = \beta$). If $\alpha^{c\Omega}, \beta^{c\Omega}$ are algebraic then $x_j \in \mathcal{L}$, and $sx_j \in \overline{\mathbb{Q}}\log\alpha_j \subset \overline{\mathbb{Q}}\mathcal{L}$. A K-relation among $2\pi i, x_1, x_2$ without the $2\pi i$ term contradicts Baker; one with it gives $0 \ne \ell \in \overline{\mathbb{Q}}\mathcal{L} \cap \overline{\mathbb{Q}}^\times\eta^*$, impossible by the analytic subgroup theorem on $\mathbb{G}_\mathrm{m}^r \times E^\natural$ [36]. (Remark 20 gives a proof that needs no such step.) □

Remark 20 (Second derivation via §5d)**.** As in Remark 15, keep $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times Q''$, Ψ'', V'' and $W'' = \ker \Psi''$ from the proof of Theorem 18: $d = 5n/2 + 1$, $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2 + 1$, $t = d - 2$. With the $3n$ lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$ one has $m = 3n$ and $\kappa \geq n$ (in fact $\kappa = n$), and (5.4) reads $3n > (2n + n + 2 - \kappa) \cdot 1$, i.e. $3n > 2n + 2$. It holds as soon as $\kappa \geq 3$, so for every admissible $n \geq 4$ (at $n = 2$ it would read $6 > 6$). For $K = \mathbb{Q}(i, \sqrt{2})$, $n = 4$: $G'' = \mathbb{G}_{\mathrm{m}}^{8} \times Q''$, $d = 11$, $t = 9$, $m = 12$, $\kappa = 4$, and (5.4) is $12 > 14 - \kappa$.

The rank. For the data of Theorem 18 it follows from the first coordinates, as in Remark 15. For Corollary 19 take $s = 1/(c\Omega) \in \overline{\mathbb{Q}}^{\times} \eta^*/\pi$, so that $s x_j = \log \alpha_j \in \mathcal{L}$, and take the data lifts toric, $(x_j e_h,\ \log \alpha_j\, e_h)$. Take the period lifts with toric part $(2\pi i D e_h, 0)$; the second coordinate $D\theta_h \cdot 2\pi i s$ of their image is then carried by the fibre coordinate σ of Q''. So σ separates the period lifts from the data lifts, and the first-block coordinates $x_j e_h$ of the data lifts are $\mathbb{Q}$-independent as soon as $\log \alpha, \log \beta$ are.

No subgroup in V''. A connected $G' \neq 0$ with $T_{G'} \subset V''$ is $T_L \times H$ with $H \in \{0, F, \pi^{-1}(B')\}$. If $H \neq 0$, then $F \subset H$, and T_F contains a vector with image $(0, 1) \notin V$. If $H = 0$, then $L \neq 0$ and, as in Remark 15, a nonzero $l \in L$ gives $s \in K$, which is false since $s \in \overline{\mathbb{Q}}^{\times}/\Omega$ and Ω is transcendental.

So this route, too, needs only the $\mathbb{Q}$-independence of $\log \alpha, \log \beta$ and no Wüstholz step. It rests on Theorem 11 as published.

By the symmetry $t \leftrightarrow 1/t$, Corollaries 14 and 19 give the four exponentials conjecture at every $t \in \overline{\mathbb{Q}}^{\times}\Omega/\pi \cup \overline{\mathbb{Q}}^{\times}\pi/\Omega$ and at every $t \in \overline{\mathbb{Q}}^{\times}\Omega \cup \overline{\mathbb{Q}}^{\times}/\Omega$, for every CM period Ω.

3.2. **The affine version and strong four exponentials.** Adding one factor $\mathbb{G}_{\mathrm{a}}$ to each coordinate of Ψ (so $G = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^2 \times E$ with coordinates (a, a', x, y, z) and $\Psi(a, a', x, y, z) = (x + a,\ y + z + a')$) gives the following statement. In the quotient $G'' = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ one has $d_0 = 2$, which does not enter the left-hand side of Theorem 9. The only new subgroups are $\mathbb{G}_{\mathrm{a}}$-lines $\{(\mu a, a)\}$ with $\mu \notin K$, which are excluded with margin n. So the count is the same as for Theorem 13; Lemma 32 applies since $5n + 2 \geq 3n + 4$. This was checked at proof-sketch level by both internal referees of Theorem 13 and by a third; the subgroup list is not written out here. The base change in K now uses Baker's inhomogeneous theorem.

Theorem 21 (Affine MG; proof sketch)**.** *With E, Ω, c, $s = c\Omega/\pi$ and K as in Theorem 13,*

$$\dim_K \{x \in K + K\mathcal{L} :\ s\,x \in K + K\mathcal{L} + K\mathcal{L}_E\} \leq 2.$$

Corollary 22 (proof sketch)**.**

(i) *For $a, b, c \in \overline{\mathbb{Q}}^{\times}$, one of $e^{a\pi^2}$, $e^{b\pi\Omega}$, $e^{c\Omega/\pi}$ is transcendental.*

(ii) *For $\alpha \in \overline{\mathbb{Q}}^{\times}$ not a root of unity, $e^{\Omega/\pi}$ and $\alpha^{\Omega/\pi}$ (any determination) are not both algebraic.*

(iii) *(Strong four exponentials at ratio $t \in \overline{\mathbb{Q}}^{\times}\Omega/\pi$.) If y_1, y_2 are linearly independent over $\overline{\mathbb{Q}}$, then y_1, y_2, ty_1, ty_2 are not all in $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. By $y \mapsto ty$ the same holds at $t \in \overline{\mathbb{Q}}^{\times}\pi/\Omega$. (The strong statement is invariant under $t \mapsto ct$, $c \in \overline{\mathbb{Q}}^{\times}$, and $\overline{\mathbb{Q}}^{\times}\Omega/\pi$ contains a real number. So this is a statement at real t, where the $\widetilde{\mathcal{L}}$-version of the conjugation argument* [30, Thm. 4.1] *applies when $y_1, y_2, \bar{y}_2$ are $\overline{\mathbb{Q}}$-independent. The new content is the remaining y.)*

Sketch. (i) Take $x_1 = 1$ ($s \cdot 1 \in \overline{\mathbb{Q}}\mathcal{L}$ if $e^{c\Omega/\pi}$ is algebraic, with s adjusted by a constant) and $x_2 = \pi^2$. With $2\pi i$ these are K-independent, because $1, \pi, \pi^2$ are $\overline{\mathbb{Q}}$-independent (Lindemann). (ii) Take $x_1 = 1$ and $x_2 = \log \alpha$; then $1, 2\pi i, \log \alpha$ are K-independent by Baker's theorem. (iii) Take K containing all the coefficients. Then $y_1, y_2 \in T_K := \{x \in K + K\mathcal{L} :\ sx \in K + K\mathcal{L} + K\mathcal{L}_E\}$,

which has dimension ≤ 2 and contains $2\pi i$. Since y_1, y_2 are K-independent, $2\pi i \in Ky_1 + Ky_2$. Then $2ic\Omega = s \cdot 2\pi i \in \widetilde{\mathcal{L}}$, which contradicts the analytic subgroup theorem ($\Omega \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$). □

3.3. **The general form, and elliptic four exponentials.** Theorems 13 and 18 are the cases $\sigma_1 = \sigma_3 = \mathbb{G}_{\mathrm{m}}$ of the following count. Let $\sigma_1, \sigma_2, \sigma_3$ be "slots": $\mathbb{G}_{\mathrm{m}}$ (period $2\pi i$), the base of a CM curve (period Ω), or the fibre of the universal vector extension of a CM curve (period η^*). In the count of Theorem 9 the first two have weight 1 per factor and a fibre slot has weight $1 + 2/n$. Write $\Lambda_K(\sigma)$ for the K-span of the corresponding logarithms ($\mathcal{L}$, $\mathcal{L}_E$, or fibre values). Let p_j be the period of σ_j, $r = p_2/p_1$, and take Ψ reading σ_1 into the first coordinate and $\sigma_2 + \sigma_3$ into the second.

Theorem 23 (General form; proof sketch)**.** *Assume:*

- *σ_1, σ_2 are not of the same kind on isogenous groups (so $r \notin \overline{\mathbb{Q}}$);*
- *if σ_2 and σ_3 are of the same kind on isogenous groups, they are merged into one slot;*
- *(allowing $r \in \overline{\mathbb{Q}}^{\times} p_2/p_1$) K contains this constant, the CM fields, and the Legendre and isogeny constants, and $[K : \mathbb{Q}] \geq 2f + 2$, where f is the number of fibre slots;*
- *σ_1, σ_3 are not both fibre slots.*

Then

$$\dim_K\{\lambda \in \Lambda_K(\sigma_1) : \ r\lambda \in \Lambda_K(\sigma_2) + \Lambda_K(\sigma_3)\} \leq 2.$$

The count is that of Theorem 13, with the threshold $n > 2f$ coming from $G' = 0$ ($5n + 2f > 4n + 4f$). An internal referee checked all 27 slot-kind triples by an exact integer case check. The load-bearing step (a bound on the data in subtori with image a K-line) was done by hand, and the subgroup lists are not written out in this paper. The excluded case is "four exponentials for quasi-logarithms": σ_1, σ_3 fibre slots of isogenous curves (for non-isogenous curves we have not written it out). It has mixed subgroups whose fibre slope μ is algebraic but possibly outside K, and it needs the $K(\mu)$-linear independence of K-independent fibre values, which we have not proved. When $\sigma_1 = \sigma_3$ is the base of a CM curve E, Theorem 23 is an elliptic analogue of four exponentials:

Corollary 24 (Elliptic four exponentials at CM slopes; proof sketch)**.** *Let E have CM, with period Ω, and let t be one of π/Ω^2, Ω'/Ω, $\eta^*(\Omega')/\Omega$, where Ω' is a period of a CM curve E' not isogenous to E, all up to $\overline{\mathbb{Q}}^{\times}$. If $u_1, u_2 \in \mathcal{L}_E$ and Ω are linearly independent over k, then tu_1 and tu_2 do not both lie in $\overline{\mathbb{Q}}\mathcal{L}_E + \overline{\mathbb{Q}}\Lambda(\sigma_2)$. Here $\Lambda(\sigma_2)$ is the set of fibre values of E (for $t = \pi/\Omega^2$), $\mathcal{L}_{E'}$ (for $t = \Omega'/\Omega$), or the set of fibre values of E' (for $t = \eta^*(\Omega')/\Omega$). In particular tu_1, tu_2 are not both elliptic logarithms of algebraic points of E.*

4. Theorem M

4.1. **First proof (the count).** Let $G = E \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$, $\Psi(e, t, a) = (e, t + a)$, $V = \mathbb{C}(\Omega, i\pi)$, $W = 0$, and $K \supseteq k$ containing the algebraic coefficients involved. Then $\alpha_K(G) = 1 + 1 + 0 = 2$; $\Psi(\Omega_K(G)) = K\Omega \times K \cdot 2\pi i$ since $K\Lambda_E = K\Omega$ for CM; so $\kappa = 1$. Condition (i): V has a nonzero rational point iff $\Omega/\pi \in \overline{\mathbb{Q}}$, which is false (Masser [17]; cf. [29, Thm. 26]; also Schneider). Theorem 8 gives $\dim_K(V \cap \Psi(\Lambda_K)) \leq 1$, and the period $(\Omega, i\pi)$ lies in it. If $i\pi u = \Omega\ell$ then $(u, \ell) \in V \cap \Psi(\Lambda_K)$, so $(u, \ell) = s(\Omega, i\pi)$ with $s \in K$. If $u \in \mathcal{L}_E$, Masser's theorem ([29, Thm. 27]: elements of $\mathcal{L}_E$ that are k-independent are $\overline{\mathbb{Q}}$-independent) gives $s \in k$. □

Controls. If E has no CM, $\alpha_K(E) = 2$ and the count ties (no conclusion). Replacing E by $\mathbb{G}_{\mathrm{m}}$ ($\Omega \to 2\pi i$) makes V rational, so (i) fails — as it must, since $u = 2\ell$ solves $i\pi u = 2\pi i\ell$. For e^{π^2} itself ($G = \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}^2$, $V = \mathbb{C}(1, i\pi)$) one finds $\kappa = 0$ and an exact tie.

4.2. **Second proof for $u \in \mathcal{L}_E$ (Laurent's identity and Wüstholz).** We treat $u \in \mathcal{L}_E$, which covers every item of Corollary 7; a version for $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ would need several points P_j and is not written out here. Let $\Lambda_E = \Omega\,\mathbb{Z}[\alpha]$ (a sublattice of finite index suffices), and write Masser's relation [17] as $\eta(\alpha\Omega) = \bar\alpha\eta(\Omega) + c\,\Omega$ with $c \in \overline{\mathbb{Q}}$. Legendre's relation gives $(\alpha-\bar\alpha)\eta(\Omega) - c\Omega = 2\pi i/\Omega$, and $f(z) = \zeta(\alpha z) - \bar\alpha\zeta(z) - cz$ is an elliptic function defined over $\overline{\mathbb{Q}}$. With the third-kind period $\lambda(u,\Omega) = \Omega\zeta(u) - \eta(\Omega)u$ one obtains, for $u \in \mathcal{L}_E$ not a pole of f,

$$\frac{2\pi i\,u}{\Omega} = \bar\alpha\,\lambda(u,\Omega) - \lambda(\alpha u,\Omega) + \Omega f(u), \qquad f(u) \in \overline{\mathbb{Q}}. \tag{1}$$

On E_i this reads $2\eta(\varpi)u = i\lambda(iu,\varpi) - \lambda(u,\varpi)$ (cf. [16, §9, p. 160], where the general CM form is derived; the display above is its case $\mu = i$). Hence $i\pi u = \Omega\ell$ with $\ell \in \widetilde{\mathcal{L}}$ is a $\overline{\mathbb{Q}}$-linear relation among 1, logarithms of algebraic numbers, Ω, $\lambda(u,\Omega)$ and $\lambda(\alpha u,\Omega)$. These are coordinates of a point of $\Lambda(\mathcal{G})$ for $\mathcal{G} = G_P \times G_{\alpha P} \times \mathbb{G}_{\mathrm{m}}^r \times \mathbb{G}_{\mathrm{a}}$, where G_P is the extension of E by $\mathbb{G}_{\mathrm{m}}$ attached to $P = \exp_E(u)$ (Serre [24]) — add a factor $E^\natural$ when algebraic constants enter. By the analytic subgroup theorem [36], the smallest algebraic subgroup B whose tangent space contains this point lies in the hyperplane defined by the relation. B surjects onto the diagonal of $E \times E$, and over it the toric part carries the extension classes $[a + b\alpha]P$ for characters (a,b); these vanish only for $a = b = 0$ when P is not torsion. So B contains both fibre tori, on which the relation has nonzero coefficients — a contradiction. Hence P is torsion, which is the trivial case. □

4.3. **The corollaries.** (M1): $u = c\varpi/\pi$, $\ell = ic$ gives $s = c/\pi \notin \overline{\mathbb{Q}}$. (M2): $\ell = i\log\alpha$ gives $\log\alpha \in \overline{\mathbb{Q}}\pi$, so $\alpha = (-1)^{-is}$ with s algebraic, a root of unity by Gel'fond–Schneider. (M3): if $e^{\pi^2} \in \overline{\mathbb{Q}}$ then $\ell = ic\pi^2 \in \widetilde{\mathcal{L}}$ and $u = c\pi\Omega$ give $s = c\pi \notin \overline{\mathbb{Q}}$. (M3′): $u = \varpi\log\alpha$, $\ell = i\pi\log\alpha$ give $\log\alpha \in \overline{\mathbb{Q}}^\times$, contradicting Hermite–Lindemann; (M3) is the case $\alpha = -1$, using $\wp(iz) = -\wp(z)$. (Mη): $\eta(\varpi) = \pi/\varpi$. On a general CM curve $\eta(\Omega) = \kappa_1\Omega + \kappa_2\pi/\Omega$ with $\kappa_2 \neq 0$, and the statement becomes $(\eta - \kappa_1\Omega)u \notin \widetilde{\mathcal{L}}$.

5. Theorem MX

5.1. **Set-up.** For a CM curve E let $E^\natural$ be its universal vector extension, with base coordinate z and conjugate-eigen fibre coordinate w as in Lemma 12. In these coordinates the periods are $(\lambda, \eta^*(\lambda))$, $\lambda \in \Lambda_E$, where $\eta^*(\lambda) = \eta(\lambda) - \kappa_1\lambda$ satisfies $\eta^*(a\lambda) = \bar a\,\eta^*(\lambda)$; the logarithms of points of $E^\natural(\overline{\mathbb{Q}})$ are $(u, \zeta^*(u) + t)$ with $u \in \mathcal{L}_E$, $t \in \overline{\mathbb{Q}}$ and $\zeta^* = \zeta - \kappa_1\,\mathrm{id}$. On E_i and E_ρ, $\kappa_1 = 0$. Since the characters $a \mapsto a, \bar a$ of k are K-independent for $K \supseteq k$,

$$\Omega_K(E^\natural) = K\lambda_0 \times K\eta^*(\lambda_0) \qquad (\Lambda_E \otimes \mathbb{Q} = k\lambda_0). \tag{2}$$

5.2. **Proof of Theorem 3.** Let $G = E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{a}}^2$, let $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ be a number field, $n = [K:\mathbb{Q}]$ (so $4 \mid n$), and let

$$\Psi(x_i, x_\rho, a_1, a_2) = \big(z(x_i) + z(x_\rho) + a_1,\ w(x_i) + w(x_\rho) + a_2\big), \qquad V = \mathbb{C}\,w_1,$$

with $w_1 = (\varpi, \beta\eta_\rho)$, $\eta_i = \eta(\varpi) = \pi/\varpi$, $\eta_\rho = \eta(\Omega_\rho) = 2\pi/(\sqrt{3}\,\Omega_\rho)$, and $w_2 = (\Omega_\rho, \beta\frac{2}{\sqrt{3}}\eta_i)$. By (2), $w_1, w_2 \in \Psi(\Omega_K(G))$ and

$$\Psi(\Omega_K(G)) = P := (K\varpi + K\Omega_\rho) \times (K\eta_i + K\eta_\rho);$$

$w_1 \parallel w_2$ because $\varpi \cdot \frac{2}{\sqrt{3}}\eta_i = \frac{2\pi}{\sqrt{3}} = \Omega_\rho\eta_\rho$, and w_1, w_2 are K-independent since $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$ (periods of non-isogenous curves are $\overline{\mathbb{Q}}$-independent [36]).

Proposition 25. $V \cap \Psi(\Lambda_K(G)) = Kw_1 + Kw_2$.

Granting Proposition 25: the point (X, Y, a, t) lies in $\Lambda(G)$ and the hypothesis says that its image lies on V, so $z(X) + z(Y) + a = x\varpi + x'\Omega_\rho$ with $x, x' \in K$. By the linear independence of 1 and elliptic logarithms on non-isogenous curves (Wüstholz [36]), $a = 0$, $z(X) = x\varpi$, $z(Y) = x'\Omega_\rho$, and by Masser's theorem [17] $x \in \mathbb{Q}(i)$, $x' \in \mathbb{Q}(\rho)$. □

Proof of Proposition 25. Suppose $v \in V \cap \Psi(\Lambda_K(G))$ with $v \notin Kw_1 + Kw_2$.

Step 1: an elementary lemma. For $L \in \{0,\ \mathbb{C}(1,0),\ \mathbb{C}(0,1)\}$ we have $V \cap (L+P) = Kw_1 + Kw_2$. Indeed, if $sw_1 \in L + P$ with $L \neq \mathbb{C}(1,0)$ then $s\varpi \in K\varpi + K\Omega_\rho$, so $s \in K + K\Omega_\rho/\varpi$ and $(\Omega_\rho/\varpi)w_1 = w_2$; if $L = \mathbb{C}(1,0)$ then $s\beta\eta_\rho \in K\eta_i + K\eta_\rho$, so $s \in K + K\eta_i/\eta_\rho$ and $(\eta_i/\eta_\rho)w_1 = (\pi/\eta_\rho, \beta\eta_i) = \frac{\sqrt{3}}{2}w_2$.

Step 2: the group. Fix a basis $\theta_1, \dots, \theta_n$ of $K/\mathbb{Q}$, let $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, and $\Phi = \Psi \circ p_\theta$. Write the coordinates of T_{G^n} as $z_i, w_i, z_\rho, w_\rho, a_1, a_2 \in \mathbb{C}^n$ and let $U \subset G^n$ be the vector subgroup $\{z_i = z_\rho = 0,\ \theta^{\mathsf{T}} a_1 = 0,\ \theta^{\mathsf{T}}(w_i + w_\rho + a_2) = 0\}$, of dimension $4n - 2$, defined over $\overline{\mathbb{Q}}$, with $T_U \subset \ker \Phi$. Then $G'' := G^n/U = \mathbb{G}_{\mathrm{a}} \times G_\ell$, where G_ℓ is the extension of $A = E_i^n \times E_\rho^n$ by $\mathbb{G}_{\mathrm{a}}$ obtained from the universal vector extension of A by pushing out along $(w_i, w_\rho) \mapsto \theta^{\mathsf{T}} w_i + \theta^{\mathsf{T}} w_\rho$. In the coordinates $s_1 = \theta^{\mathsf{T}} a_1$, z_i, z_ρ, $s_2 = \theta^{\mathsf{T}}(w_i + w_\rho + a_2)$ the map induced by Φ is

$$\Psi''(s_1, z_i, z_\rho, s_2) = \big(\theta^{\mathsf{T}} z_i + \theta^{\mathsf{T}} z_\rho + s_1,\ s_2\big).$$

We apply [21, Thm. 4.1] (Theorem 9) to G'' with $d = 2n + 2$, $d_0 = 1$ ($\pi_0 = s_1$), $d_1 = 0$, $G_2 = G_\ell$, $d_2 = 2n + 1$; $V'' = \Psi''^{-1}(V)$, a hyperplane, so $N = 2n + 1$; $W'' = \ker \Psi''$, rational of dimension $2n$; and Y'' generated by the images of $3n$ elements of $\Lambda(G^n)$ whose Φ-images are $D\theta_h w_1$, $D\theta_h w_2$ (taken in $\Omega(G^n)$, D a common denominator) and $\theta_h v$, $h = 1, \dots, n$. Since U is a vector group, $\Omega(G'')$ is the image of $\Omega(G^n)$, so $\Psi''(\Omega(G'')) \subset P$. By Step 1 with $L = 0$, $\kappa = \mathrm{rk}(Y'' \cap \Omega(G'')) = 2n$, and $\Gamma = \exp Y''$ has rank n. The inequality of Theorem 9 becomes

$$\lambda + 2\delta_2 \leq (\delta - \tau)(2n + 2), \tag{3}$$

and, with $r = \dim \Psi''(T_{G'})$, one has $\tau = \delta + r - 2$; so an obstruction must have $r \leq 1$.

Step 3: the subgroups of G''. Abelian subvarieties of A are $B = B_i \times B_\rho$ (E_i, E_ρ are non-isogenous), and $B_i \subset E_i^n$ corresponds to a $\mathbb{Q}(i)$-subspace $S_i \subset \mathbb{Q}(i)^n$ with $T_{B_i} = \mathbb{C}\,\sigma(S_i)$, $\dim B_i = \dim S_i$; likewise for E_ρ. With ψ, ϕ as in Lemma 10 (for $k = \mathbb{Q}(i)$ and for $k = \mathbb{Q}(\rho)$), $\ker \psi \cap \ker \phi = 0$ and each kernel has k-dimension $n/2$. Because $[a]$ acts on $T(E^\natural)$ by $\mathrm{diag}(a, \bar{a})$ and $\mathrm{Hom}(E^\natural, \mathbb{G}_{\mathrm{a}}) = 0$ (the universal vector extension is anti-affine; [22]), the image in G_ℓ of the universal extension of B is a splitting $s(B) \cong B$ (with tangent space $T_B \times 0$) if $\phi(S_i) = \phi(S_\rho) = 0$, and is $\pi^{-1}(B)$, a non-split extension, otherwise (a splitting would give a nonzero homomorphism from a quotient of $(E_i^\natural)^{b_i} \times (E_\rho^\natural)^{b_\rho}$ to $\mathbb{G}_{\mathrm{a}}$). A connected subgroup over B either contains the fibre or is a splitting, and a splitting is unique since $\mathrm{Hom}(B, \mathbb{G}_{\mathrm{a}}) = 0$. It follows that the connected algebraic subgroups of $G'' = \mathbb{G}_{\mathrm{a}} \times G_\ell$ are exactly

type	G'	condition	$\Psi''(T_{G'})$
A	$\mathbb{G}_{\mathrm{a}} \times \pi^{-1}(B)$	—	$\mathbb{C}^2$
B	$\mathbb{G}_{\mathrm{a}} \times s(B)$	$\phi(S) = 0$	$\mathbb{C}(1,0)$
C	$0 \times \pi^{-1}(B)$	—	$\mathbb{C}(0,1)$ if $\psi(S) = 0$, else $\mathbb{C}^2$
D	$0 \times s(B)$	$\phi(S) = 0$	0 if $B = 0$, else $\mathbb{C}(1,0)$
E	$\{(\mu\,\mathrm{pr}_F(g), g) : g \in s(B) \times F\}$, $\mu \neq 0$	$\phi(S) = 0$	$\mathbb{C}(\mu, 1)$ if $B = 0$, else $\mathbb{C}^2$

(with $\phi(S) = 0$ meaning $\phi(S_i) = \phi(S_\rho) = 0$, and similarly for ψ). In type E, $F \cong \mathbb{G}_{\mathrm{a}}$ is the fibre of $G_\ell \to A$ (coordinate s_2), $\pi^{-1}(B) = s(B) \times F$, and pr_F is the projection to F; so G' is the graph of $\mu\,\mathrm{pr}_F$ into the first factor $\mathbb{G}_{\mathrm{a}}$ (coordinate s_1), and for $B = 0$ it is $\{s_1 = \mu s_2\}$. In every case allowed by the conditions, $b = \dim B \leq n$.

Step 4: no obstruction. If $\Psi''(T_{G'}) \in \{0, \mathbb{C}(1,0), \mathbb{C}(0,1)\}$ then $\lambda = n$: if $\exp(\sum c_h \tilde{y}_h) \in G'$ for the lifts $\tilde{y}_h$ of $\theta_h v$, then $xv \in L + P$ with $x = \sum c_h \theta_h \in K$, so $x = 0$ by Step 1. With $\delta_0 = 0$ for types A, B, E and $\delta_0 = 1$ for C, D, inequality (3) reads: $5n + 2 \le 4n + 4$ for $G' = 0$; $5n + 2 - 2b \le 2n + 2$ for types B and D ($B \neq 0$); $5n - 2b \le 2n + 2$ for type C with $r = 1$; and $\lambda + 4n + 2 \le 2n + 2$ for type E with $B = 0$. Since $b \le n$, none of these holds for $n \ge 3$, and types with $r = 2$ fail $\delta > \tau$. So no connected $G' \neq G''$ satisfies the conclusion of Theorem 9, a contradiction. □

Remark 26 (The count)**.** Theorem 8 predicts Proposition 25: with $\kappa = 2$ and $\alpha_K(G) = 2(2 + 2/n)$ (Lemma 12) it gives $\dim_K(V \cap \Psi(\Lambda_K(G))) \le 2 + 4/n$, which is < 3 only for $n > 4$; with the weight $\alpha(G'')/n = (4n + 2)/n$ one gets $2 + 2/n < 3$ for $n \ge 3$, which is the threshold of Step 4.

A second derivation of Proposition 25 applies Theorem 11 to (G'', V'', W'', Y''): $d = 2n + 2$, $d_0 = 1$, $d_1 = 0$, $d_2 = 2n + 1$, $t = \dim W'' = 2n$, $m = 3n$, $\kappa = 2n$, and (5.4) reads $3n > (4n + 2 - 2n) \cdot 1$, i.e. $3n > 2n + 2$ (for $K = \mathbb{Q}(i, \sqrt{3})$, $n = 4$: $12 > 18 - \kappa = 10$). It remains to see that no connected $G' \neq 0$ has $T_{G'} \subset V''$. First, $T_{G'} \not\subset \ker \Psi''$. If G' meets the kernel $\mathbb{G}_a \times F$ of $G'' \to A$ in a subgroup of positive dimension, $T_{G'}$ contains a vector $(c, 0, 0, d) \neq 0$, whose image is (c, d). Otherwise G' is finite over its image B in A, and $B \neq 0$. Then G' is an abelian variety, so $G' \subset 0 \times G_\ell$ ($\mathrm{Hom}(B, \mathbb{G}_a) = 0$) is a splitting $s(B)$; by Step 3 this forces $\phi(S) = 0$, and then $\psi \neq 0$ on S (Lemma 10), so $\Psi''(T_{G'}) = \mathbb{C}(1, 0)$. Since $T_{G'}$ is $\overline{\mathbb{Q}}$-rational, it contains a $\overline{\mathbb{Q}}$-rational x with $\Psi''(x) \neq 0$. If $T_{G'} \subset V''$, then $\Psi''(x) \in V$ is a nonzero $\overline{\mathbb{Q}}$-rational point of $V = \mathbb{C}w_1$, so $\varpi/(\beta\eta_\rho) \in \overline{\mathbb{Q}}$. This contradicts $\varpi/\eta_\rho \notin \overline{\mathbb{Q}}$, which is Wüstholz's theorem [36] (a period of E_i and a quasi-period of E_ρ are $\overline{\mathbb{Q}}$-linearly independent). The same exclusion can be read off the table in Step 3. Unlike Step 4, this derivation uses $\varpi/\eta_\rho \notin \overline{\mathbb{Q}}$, and it rests on Theorem 11 as published.

Corollary 4 is the case $X = (u, \zeta_i(u))$, $Y = 0$ (the logarithm of the origin), resp. $X = 0$; the value 0 of the ratio is excluded separately, since $1, u, \zeta(u)$ are $\overline{\mathbb{Q}}$-linearly independent for non-torsion u on a CM curve [15]; it also follows from [36] applied on $E^\natural \times \mathbb{G}_a$. Note that the base and fibre coordinates of a single logarithm cannot be chosen independently: $w(X) - \zeta^*(z(X)) \in \overline{\mathbb{Q}}$.

Remark 27 (Other pairs of curves)**.** Let E, E' be non-isogenous CM curves with lattices $\omega\mathbb{Z}[\alpha]$, $\omega'\mathbb{Z}[\alpha']$ (up to finite index), so that $(\alpha - \bar{\alpha})\,\omega\,\eta^*(\omega) = 2\pi i = (\alpha' - \bar{\alpha}')\,\omega'\eta^*(\omega')$. With $V = \mathbb{C}(\omega, \beta\eta^*(\omega'))$ the vectors $w_1 = (\omega, \beta\eta^*(\omega'))$ and $w_2 = \big(\omega', \beta\frac{\alpha - \bar{\alpha}}{\alpha' - \bar{\alpha}'}\eta^*(\omega)\big)$ are parallel, and the same proof gives: if $X \in \Lambda(E^\natural)$, $Y \in \Lambda(E'^\natural)$ and $(z(X) + z(Y) + a)\beta\eta^*(\omega') = (w(X) + w(Y) + t)\omega$, then $a = 0$ and $z(X) \in k\omega$, $z(Y) \in k'\omega'$.

Remark 28. By the explicit formulas for ϖ and Ω_ρ, $\varpi\Omega_\rho/\pi \in \overline{\mathbb{Q}}^\times \cdot \Gamma(1/4)^2\Gamma(1/3)^3/\pi^{5/2}$, and $\varpi/\eta_\rho = \frac{\sqrt{3}}{2\pi}\varpi\Omega_\rho$. If $\varpi/\eta_\rho = c \in \overline{\mathbb{Q}}$, then $X = Y = 0$, $a = 1$, $t = \beta/c$ satisfy the hypothesis of Theorem 3 with $a \neq 0$; so Corollary 5 is the case $X = Y = 0$. It also follows directly from Wüstholz's theorem [36] (a period of E_i and a quasi-period of E_ρ are $\overline{\mathbb{Q}}$-linearly independent).

5.3. **Controls.** (1) If either curve has no CM there is no relation making two periods parallel, so κ drops to n in Step 2 and $G' = 0$ satisfies the inequality of Theorem 9 ($5n + 2 \le 6n + 4$): no conclusion. The same happens if only one Legendre relation is used. If the fibre action were $\mathrm{diag}(a, a)$ instead of $\mathrm{diag}(a, \bar{a})$, subgroups of type D with image 0 would satisfy it; and for a rational line $V = \mathbb{C}(\mu, 1)$ with true data on it ($\kappa = 0$) the type E line does. So the conjugate character, the second Legendre relation and the irrationality of V are all used. (2) With the same curve twice, the antidiagonal of $E^\natural \times E^\natural$ lies in $\ker \Psi$ and the count collapses to the one-curve cell, which only gives $(\zeta(u) + t)/(u + a) \neq \pi/\varpi^2$ — a weak form of Chudnovsky's theorem [5]. (3) Data on the period plane satisfy the relation only trivially.

5.4. **A twisted variant.** Exchanging the roles of base and fibre on E_ρ gives a second statement of the same kind. Let $G = E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{a}}^2$ and

$$\Psi'(X_i, X_\rho, a_1, a_2) = \big(z(X_i)+w(X_\rho)+a_1,\ w(X_i)+z(X_\rho)+a_2\big), \qquad V' = \mathbb{C}\, w_1', \quad w_1' = (\varpi, \beta\Omega_\rho).$$

With $w_2' = (\eta_\rho, \frac{2\beta}{\sqrt{3}}\eta_i)$ one has $w_2' = (\eta_\rho/\varpi)\, w_1'$ by the two Legendre relations, and for $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ both lie in $\Psi'(\Omega_K(G)) = (K\varpi + K\eta_\rho) \times (K\eta_i + K\Omega_\rho)$.

Theorem 29 (Theorem MX′)**.** *Let $X \in \Lambda(E_i^\natural)$, $Y \in \Lambda(E_\rho^\natural)$, $a, b \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^\times$. If*

$$\big(z(X) + w(Y) + a\big) \cdot \beta\, \Omega_\rho = \big(w(X) + z(Y) + b\big) \cdot \varpi,$$

then $z(X) = z(Y) = 0$ and both sides vanish.

Proof (sketch). Proposition D′: for $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ of degree $n \geq 8$, $V' \cap \Psi'(\Lambda_K(G)) = Kw_1' + Kw_2'$. With θ, p_θ and $\Phi' = \Psi' \circ p_\theta$ as before, a canonical lift $B^\natural$ of an abelian subvariety $B_{S_i} \times B_{S_\rho}$ has tangent space in $\ker \Phi'$ only if $\psi(S_\bullet) = \phi(S_\bullet) = 0$, i.e. $B = 0$ (Lemma 10). So the largest connected subgroup U' with $T_{U'} \subset \ker \Phi'$ is the vector subgroup $\{\theta^{\mathsf{T}}(w_\rho + a_1) = 0,\ \theta^{\mathsf{T}}(w_i + a_2) = 0\}$, and $G'' = G^n/U'$ is the pushout of $(E_i^n \times E_\rho^n)^\natural$ along $(w_i, w_\rho) \mapsto (\theta^{\mathsf{T}} w_i, \theta^{\mathsf{T}} w_\rho)$, an extension of $E_i^n \times E_\rho^n$ by $\mathbb{G}_{\mathrm{a}}^2$ of dimension $2n+2$. It is a quotient of an anti-affine group, so $\mathrm{Hom}(G'', \mathbb{G}_{\mathrm{a}}) = \mathrm{Hom}(G'', \mathbb{G}_{\mathrm{m}}) = 0$: $d_0 = d_1 = 0$, $d_2 = 2n+2$. With the lifts of $D\theta_h w_1'$, $D\theta_h w_2'$ and $\theta_h v$ for a hypothetical $v \notin Kw_1' + Kw_2'$ one has $m = 3n$ and $\kappa = 2n$. The κ-refined Theorem 1.1 of [21, §5d] (Theorem 11) applies: $m > (d_1 + 2d_2 - \kappa)(d - 1 - t)$ reads $3n > 2n + 4$, i.e. $n > 4$, hence $n \geq 8$ since $4 \mid n$. Its remaining hypothesis holds because $\Psi''(T_{G'})$ would be a $\overline{\mathbb{Q}}$-rational subspace of V', which has no $\overline{\mathbb{Q}}$-rational point ($\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$), so $T_{G'} \subset \ker \Psi''$ and $G' = 0$ by the maximality of U'. (A full case check against Theorem 9, with the analogue of Step 1 of the proof of Theorem 3, gives the same threshold.)

Conclusion. By D′, $z(X) + w(Y) + a = x\varpi + x'\eta_\rho$ and $w(X) + z(Y) + b = x\beta\Omega_\rho + x'\frac{2\beta}{\sqrt{3}}\eta_i$ with $x, x' \in K$. The analytic subgroup theorem gives $z(X) = p\varpi$, $z(Y) = q\Omega_\rho$ with $p \in \mathbb{Q}(i)$, $q \in \mathbb{Q}(\rho)$; then $\zeta_i(p\varpi) \in \bar{p}\,\eta_i + \overline{\mathbb{Q}}$ and $\zeta_\rho(q\Omega_\rho) \in \bar{q}\,\eta_\rho + \overline{\mathbb{Q}}$, and the $\overline{\mathbb{Q}}$-independence of $1, \varpi, \eta_\rho$ and of $1, \eta_i, \Omega_\rho$ gives $x = p$, $x' = \bar{q}$, $q = p\beta$, $\bar{p} = \frac{2\beta}{\sqrt{3}}\bar{q}$, and vanishing constant terms. If $p \neq 0$ this gives $\sqrt{3}|p|^2 = 2|q|^2$ with $|p|^2, |q|^2 \in \mathbb{Q}_{>0}$, which is impossible. □

Corollary 30. *For every nonzero $u \in \mathcal{L}_{E_i}$ (off the lattice) and $a, b \in \overline{\mathbb{Q}}$: $(\zeta_i(u) + b)/(u + a) \notin \overline{\mathbb{Q}}^\times \cdot \Omega_\rho/\varpi = \overline{\mathbb{Q}}^\times \cdot \Gamma(1/3)^3/(\sqrt{\pi}\,\Gamma(1/4)^2)$.*

Together with Corollary 4, the ratio $(\zeta_i(u) + t)/(u + a)$ avoids both $\overline{\mathbb{Q}}\,\pi/(\varpi\Omega_\rho)$ and $\overline{\mathbb{Q}}^\times \Omega_\rho/\varpi$.

5.5. **Two degree-two period statements.** Propositions 25 and D′ also apply to data that are periods plus algebraic constants. This gives statements about products of periods of the two curves.

Corollary 31. (i) *The numbers $\pi,\ \varpi,\ \Omega_\rho,\ \varpi\Omega_\rho$ are linearly independent over $\overline{\mathbb{Q}}$; equivalently, so are $\pi, \eta_i, \eta_\rho, \eta_i\eta_\rho$.*

(ii) *The numbers $1,\ \varpi,\ \Omega_\rho,\ \varpi\Omega_\rho/\pi$ are linearly independent over $\overline{\mathbb{Q}}$; equivalently, so are $\pi, \varpi, \eta_\rho, \varpi\eta_\rho$.*

Proof. The equivalences follow by multiplying a relation by $\eta_i\eta_\rho/\pi$, resp. by η_ρ, and using $\varpi\eta_i = \pi$, $\Omega_\rho\eta_\rho = 2\pi/\sqrt{3}$. We use that $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are linearly independent over $\overline{\mathbb{Q}}$ (analytic subgroup theorem [36] on $E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$).

(i) A relation with zero coefficient on $\varpi\Omega_\rho$ is excluded by that independence, so suppose $\pi = A\varpi\Omega_\rho + B\varpi + C\Omega_\rho$ with $A \neq 0$. Take $\beta = A$, $X = (\varpi, \eta_i) \in \Lambda(E_i^\natural)$, $Y = 0$, $a = C/A$,

$b = -B$ in Theorem 29: the left side is $(\varpi + C/A)A\Omega_\rho = \pi - B\varpi$ and the right side is $(\eta_i - B)\varpi = \pi - B\varpi$. So the hypothesis holds, and Theorem 29 gives $z(X) = 0$, whereas $z(X) = \varpi$.

(ii) Suppose $\varpi\Omega_\rho/\pi = a + b\varpi + c\Omega_\rho$ (a relation with zero left-hand coefficient is excluded by the independence above). If $c = 0$, dividing by ϖ and using $\pi/\varpi = \eta_i$, $\Omega_\rho = 2\pi/(\sqrt{3}\eta_\rho)$ gives $\Omega_\rho = a\eta_i + b\pi$. If $b = 0$, multiplying by η_ρ gives $\frac{2}{\sqrt{3}}\varpi = a\eta_\rho + \frac{2c}{\sqrt{3}}\pi$. Both contradict the independence above, so $b, c \neq 0$. Put $t = -1/c$, $\beta = -\sqrt{3}\,b/(2c)$, $a_1 = a/b$, and take $X = (\varpi, \eta_i + t) \in \Lambda(E_i^\natural)$, $Y = 0$. Then $v = (\varpi + a_1, \eta_i + t) \in \Psi(\Lambda(G))$ lies on $V = \mathbb{C}(\varpi, \beta\eta_\rho)$: expanding $(\varpi + a_1)\beta\eta_\rho = (\eta_i + t)\varpi$ with $\varpi\eta_i = \pi$ and $\eta_\rho = 2\pi/(\sqrt{3}\Omega_\rho)$ gives exactly the assumed relation. Proposition 25 gives $v = x(\varpi, \beta\eta_\rho) + x'(\Omega_\rho, \frac{2\beta}{\sqrt{3}}\eta_i)$ with $x, x' \in K$. The first coordinate and the independence of $1, \varpi, \Omega_\rho$ give $a_1 = 0$, $x = 1$, $x' = 0$; then $\eta_i + t = \beta\eta_\rho$, contradicting the independence of $1, \eta_i, \eta_\rho$. □

Both statements are genuinely of degree two: rewriting with the Legendre relations always leaves a product of periods of the two different curves, so they are not consequences of the analytic subgroup theorem. They are predicted by the period conjecture (by Chowla–Selberg [7] they concern π, $\Gamma(1/4)$, $\Gamma(1/3)$, whose algebraic independence is open), and they do not follow from known algebraic independence results. They have the shape of a nonvanishing 2×2 determinant of periods, like the theorem of Barré-Sirieix, Diaz, Gramain and Philibert [2] in the form [29, Cor. 41]. We found them nowhere in print; they are the most delicate claims of the paper and we flag them for an expert check. Neither statement contains the constant 1 together with $\varpi\Omega_\rho$, so they do not give $\varpi\Omega_\rho \notin \overline{\mathbb{Q}}$. For E_i, E_ρ, that statement is the CM case of Waldschmidt's conjecture [33, slide 2] that $\omega_1\omega_2^* - \omega_1^*\omega_2$ is transcendental for non-isogenous curves, since $\varpi \cdot \rho\Omega_\rho - \Omega_\rho \cdot i\varpi = (\rho - i)\varpi\Omega_\rho$. We do not claim it.

5.6. **The abelian-surface analogue.** A simple CM abelian surface A has two eigen-pairs of periods with $P_1P_4,\ P_2P_3 \in \pi\,\overline{\mathbb{Q}}^\times$ (Riemann's bilinear relations). For $A = \mathrm{Jac}(y^2 = x^5 + 1)$, $P_a = (1 - \zeta_5^a)B(\frac{a}{5}, \frac{1}{2} - \frac{a}{5})/5$, and the reflection formula gives these relations exactly. With $G = A^\natural \times \mathbb{G}_{\mathrm{a}}^2$, $\Psi = (z_1 + z_2 + a_1,\ w_3 + w_4 + a_2)$ and $V = \mathbb{C}(P_1, \beta P_3)$, the same argument (with $\alpha_K(A^\natural) = 4 + 4/n$) shows: for non-torsion $P \in A(\overline{\mathbb{Q}})$ with logarithm v, $(z_1(v) + z_2(v) + a)\beta P_3 \neq (w_3(v) + w_4(v) + t)P_1$. We state this only as a proof sketch; the steps not written out are: the computation $\alpha_K(A^\natural) = 4 + 4/n$ (no abelian part in H, since the four coordinates carry four distinct characters of $\mathbb{Q}(\zeta_5)$); condition (i) via Wüstholz on $A^\natural$, which uses that $A^\natural$ has no proper algebraic subgroup surjecting onto A; the description of $\Lambda(A^\natural)$; and the final deduction that v is torsion. This sketch uses Theorem 8. The direct argument of §5 should carry over with the same numbers ($G'' = \mathbb{G}_{\mathrm{a}} \times G_\ell$ of dimension $2n+2$, abelian subvarieties given by $\mathbb{Q}(\zeta_5)$-subspaces and four conditions from the four embeddings), but we have not written it out.

6. Single-partner limits for e^{π^2}

The three exponentials conjecture asks that $\lambda_1\lambda_2 = \gamma\lambda_3$ with $\lambda_j \in \mathcal{L}$, $\gamma \in \overline{\mathbb{Q}}^\times$ force $\lambda_3 = 0$; $e^{\pi^2} \notin \overline{\mathbb{Q}}$ is the instance $\lambda_1 = \lambda_2 = i\pi$. In the count above, a configuration of rank-one shape gives a theorem only if κ exceeds what the true data need. For lines with a single coordinate per factor, $\kappa \leq 1$, and one CM factor supplies exactly one unit of weight. On the period side, Laurent's identity (1) and its abelian analogue show that these configurations are linearisable: multiplying by π turns them into linear relations among 1-periods, decided by the analytic subgroup theorem.

To get $\kappa = 2$ on a line one needs two independent periods with parallel images, i.e. a true *homogeneous* degree-two relation $p_ap_d = p_bp_c$ among periods. The CM Legendre relations

$(\alpha-\bar\alpha)\,\omega\,\eta^*(\omega)=2\pi i$ of two non-isogenous CM curves provide one; so do two eigen-pairs of a CM abelian variety. The multiplicative group cannot take part: its only period is $2\pi i$, and the corresponding relation would pair $2\pi i$ with 1, which is not a period (for $E^\natural\times\mathbb{G}_{\mathrm{m}}$ with $\mathbb{G}_{\mathrm{m}}$ added to either coordinate, parallel periods force a $\overline{\mathbb{Q}}$-linear relation among $1,\omega,\eta^*$, so $\kappa\le 1$). Under the hypothesis $e^{\pi^2}\in\overline{\mathbb{Q}}$ the number π^2 is a logarithm, not a period, so it adds data, not κ. This is why Theorem 3 exists while the same count cannot reach e^{π^2}. The $\mathbb{G}_{\mathrm{m}}$-analogue of Theorem 3 (replace Ω_ρ by $i\pi$ and the E_i-data by logarithms) is $i\pi\cdot\log\beta=\gamma\log\alpha$, an open case of the conjecture.

With two partners the mechanism does reach e^{π^2}: Corollary 16 (one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$) uses π^2 and π^2/Ω as data and the period $(2\pi i, 2ic\Omega)$ for κ. In the configurations we enumerated (lines in $\mathbb{C}^2$, at most three factors per coordinate; not refereed), no single-partner disjunction "one of e^{π^2}, X" appears beyond (M3) and the known strong-six partners $e^{c/\pi}$, $e^{c\pi^3}$ (from the strong six exponentials theorem [19, Cor. 2] with $x=(1,\pi)$ and $y=(1/\pi,1,\pi)$, resp. $y=(1,\pi,\pi^2)$).

Within the monomial model (heuristic; not refereed), a slack count explains this (proof sketch). On a line $V=\mathbb{C}(1,r)$, each slot contributes monomials: its period, plus the constant 1 for $\mathbb{G}_{\mathrm{a}}$ and for the fibre. So $\mathbb{G}_{\mathrm{a}}$ gives $\{1\}$, $\mathbb{G}_{\mathrm{m}}$ gives $\{\pi\}$, the base of E gives $\{\Omega\}$, and the fibre of $E^\natural$ gives $\{\pi/\Omega,1\}$. Their weights in the count are 0, 1, 1, $1+2/n$, and the true relations are homogeneous for the grading $\deg\pi=2$, $\deg(\text{1-period})=1$. A line count is then a matching between the monomials of the two coordinates. The hypothesis $e^{\pi^2}\in\overline{\mathbb{Q}}$ adds the monomial π^2 (degree 4) to the $\mathbb{G}_{\mathrm{m}}$-slots, and a partner hypothesis adds one monomial m_X. A single-partner disjunction needs the partner to sit opposite π^2, so $m_X=\pi^2 r^{\pm1}$, or the e^{π^2}-edges alone to fill the slack. Running through the cases leaves exactly $m_X=\pi\cdot p$ with p a period of the slot carrying the partner, i.e. $\wp(c\pi\Omega)$ (the case (M3)), together with the strong-six partners $e^{c/\pi}$ and $e^{c\pi^3}$. An exhaustive enumeration in a bounded range (at most four slots per coordinate) confirms this. On lines of type MX the slope has degree 0, so a $\mathbb{G}_{\mathrm{m}}$-slot can never be matched; e^{π^2} can enter such a count only together with a second hypothesis.

7. Consistency checks

The main results are consequences of a published theorem (completed by Lemma 32 in the case used) applied to explicit groups, so an error would most likely sit in the bookkeeping. We checked that the same bookkeeping does not prove false statements, and that it gives no conclusion in the control configurations below, where it should not.

- *Rational lines.* If V contains a $\overline{\mathbb{Q}}$-rational point, the statement is false (e.g. $\mathbb{G}_{\mathrm{m}}$ in place of E, $s=2$, $x=\log 2$). In each configuration we checked (MG, MG$^\natural$, MX) the count finds an obstruction, typically a subtorus or a $\mathbb{G}_{\mathrm{a}}$-line; for MG and MG$^\natural$ it occurs at equality in the inequality of Theorem 9.
- *No CM, wrong field, one Legendre relation, wrong coordinates.* An obstruction appears (the trivial subgroup, or a type-D subgroup) in each of these cases: a non-CM curve; $k\not\subset K$; only one Legendre relation on an MX-type line; for MX, $\mathrm{diag}(a,a)$ in place of $\mathrm{diag}(a,\bar a)$ on $T(E^\natural)$; for MG$^\natural$, a non-eigen fibre coordinate. So no conclusion is claimed in these cases. (For MG$^\natural$ the conjugation itself is not load-bearing; what matters is that Ψ reads an eigen-coordinate.)
- *Pure periods.* Data consisting of periods alone are true, so they test the period term by itself. In all the configurations we built (semi-abelian G with V a hyperplane containing a corank-one sublattice of periods, $\mathbb{G}_{\mathrm{m}}^{d_1}\times E^g$, the groups of §§3, 5), an obstruction exists.
- *Sharpness of the period term.* For the graph of complex multiplication in $E\times E$ ($\kappa=2$), the diagonal in $\mathbb{G}_{\mathrm{m}}^2$ ($\kappa=1$), and the Weil-restricted graph of complex multiplication with

maximal $\kappa = n$, the obstruction occurs exactly at equality. A period term larger by one would prove false statements there. The applications in §§3, 5 use $\kappa = \mathrm{rk}(V'' \cap \ker \exp)$ exactly. This is calibration evidence, not a proof of [21, Prop. 6.1].

- $\mathbb{G}_{\mathrm{m}}$ *alone* [proved]. For $G = \mathbb{G}_{\mathrm{m}}^r$ the period space $\Psi(\Omega_K)$ is $2\pi i$ times a K-rational subspace. So a line $V = \mathbb{C}(1, s)$ meets it only if $s \in K$, where the statement is false and the method is silent. With a logarithm in place of the elliptic period ($\kappa = 0$), the same count reproduces the six exponentials theorem (a calibration).
- *Special exponents.* Values of t that make one power algebraic ($2^t = 3$, $2^t = 1$) are not known to lie outside $\overline{\mathbb{Q}}^\times \Omega/\pi$. They lie inside only under degree-two relations such as $\pi \log 3 \in \overline{\mathbb{Q}}\,\Omega \log 2$ or $\Omega \log 2 \in \overline{\mathbb{Q}}\pi^2$, which are conjecturally false. If they did lie inside, the resulting consequences of Corollary 14 would be conjecturally true and would contradict no known result.

8. Status and sources not consulted

Theorems 13, 18, 3 and 29 (hence Theorem 1 and the corollaries) rest on the following.

- Theorem 9 [21, Thm. 4.1]. Its proof [21, §§6–7] uses:
  - the auxiliary function with periods and the κ-saving [26, Prop. 2.1, 2.4];
  - multiplicities along W (Lemma 32; a proof sketch, checked by two independent internal referees, not peer-reviewed). Its published inputs are [26, Prop. 2.1, Lemma 2.2], Serre's embedding and height bound [24], Bertrand's [25, App. I, Lemme 7 and its Corollaire, with Lemme 1] (cited in [21, p. 396] as "Lemma 7"), and the lower bound of [26, p. 640];
  - Philippon's zero estimate [11];
  - a complete system of addition laws of one fixed bidegree for the projective embedding of G_2. This exists for any quasi-projective commutative group variety by an elementary argument (Remark 33); for abelian varieties see also Lange–Ruppert [9]. (Serre's appendix [24] gives the compactification with its G-action, but not addition laws; cf. also [8], which we have not checked.)
- For the sketch of Theorem 29 and the second derivations (Remarks 15, 20, 26): Theorem 11, as published in [21, §5d].
- Legendre's relation; Wüstholz's analytic subgroup theorem [36]; Masser [17]; Baker; the transcendence of Ω and of Ω/π (Schneider, Masser [17]).
- Standard facts on CM abelian varieties and universal vector extensions.

For $u \in \mathcal{L}_E$ (all of Corollary 7), Theorem 6 follows from Laurent's identity and the analytic subgroup theorem; for $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ it rests on Statement R (in the hyperplane case, which is Theorem 11 in substance). Extensions to CM abelian varieties (four exponentials at Beta values such as $\Gamma(1/5)^2/\Gamma(2/5)$) and a mechanism using first-kind period relations of CM abelian varieties, as well as strong four exponentials at all CM slopes $\overline{\mathbb{Q}}^\times\{\Omega^{\pm 1}, (\Omega/\pi)^{\pm 1}\}$, are treated separately. What most needs an expert read: Lemma 32 (the one step of [21, Prop. 6.1] that is only indicated in print); the subgroup classifications in §§3 and 5; and the degree-two statements of Corollary 31, which say more about π, $\Gamma(1/4)$, $\Gamma(1/3)$ than any known algebraic-independence result. Corollary 31(i) rests on Theorem 29, whose write-up has had one internal check. The claim that MX is not a linearisation, and the discussion in §6, are heuristic (they use the period conjecture). Sources we could not consult, any of which might contain these statements: Bertrand [14] (the source of the K-structure used in [20]); Waldschmidt [32] (the general form of Theorem 9); Baker–Wüstholz [13, §6.2]; Bertrand–Laurent [3]; Faltings–Wüstholz [8]; Roy [18]; MathSciNet citation searches. Numerical values above were computed with mpmath at 50–60 digits; they illustrate and prove nothing.

Appendix A. Multiplicities along W in the auxiliary function

We use the notation of [21, §6]. $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ over a number field K_0 with $d_2 \geq 1$ and G_2 embedded in $\mathbb{P}_M$ by Serre's embedding (coordinates $\psi_0, \dots, \psi_M$, entire of order ≤ 2; $\xi = \psi/\psi_0$). $V \subset T_G$ has dimension $N < d$, $W \subset V$ is K_0-rational of dimension t, $y_1, \dots, y_m \in V$ with $\exp_G(y_j) \in G(K_0)$, $Y(S) = \{\sum h_j y_j : 0 \leq h_j \leq S\}$, $0 \leq \kappa \leq \mathrm{rk}_{\mathbb{Z}}(V \cap \ker \exp_G)$, and $\Delta^{d-N} = C\, S^{d_1+2d_2-\kappa}(\log S)^{bd_0}$, $T(\log S)^a = D_0(\log S)^b = D_1 S = D_2 S^2 = \Delta$.

Lemma 32. *Assume $2N \geq d_1 + 2d_0 + \kappa$, $d_2 \geq 1$, $a, b \geq 1$. For ν large and $S \geq S_0(\nu)$ there is a polynomial $P_S \in K_0[U, V, W]$ of degree $\leq D_0$ in U, $\leq D_1$ in V, homogeneous of degree $\leq D_2$ in W, not vanishing identically on G, which vanishes with multiplicity $\geq T$ along W at every point of $\exp_G(Y(S))$. Hence* [21, Prop. 6.1] *holds under these assumptions.*

Sketch. Step 1 is [26, Prop. 2.1 and p. 640] unchanged. With $r = \nu^2 S$, $R = \nu^3 S$, it gives a nonzero $Q_S \in \mathbb{Z}[U, V, W]$ of degrees $\leq D_i/\nu$ and coefficients $\leq e^{\Delta/\nu}$ with $|Q_S(\lambda, \mu, \psi)(z)| \leq e^{-\nu^6\Delta}$ for $z \in V$, $|z| \leq \nu^2 S$. The hypothesis $2N \geq d_1 + 2d_0 + \kappa$ is the condition $U \geq r^2 \log(R/r)$ of Prop. 2.1 (printed with $-\kappa$ on p. 641, a misprint). No condition along W enters the linear system, so the saving r^κ in the hypothesis of [26, Prop. 2.1, p. 633] is untouched. We also use the lower bound $\max_i |\psi_i(t+z)| \geq e^{-c(\nu S)^2}$ for $t \in Y(\nu S) \cap \mathcal{T}$ and $|z| \leq \rho$, where $Y = \mathcal{T} + F$ with F finite ([26, p. 640], [25, Lemme 1.2.2]; the ψ_i have no common zero [24, Prop. 8]). In Step 2 it bounds the scalar $\mu_j(z)$ in $E^{(j)}(\xi(\gamma), \xi(z)) = \mu_j(z)\psi(t+z)$. (The polynomial variables U, V, W below are not the subspaces V, W.)

Step 2 (derivatives). Fix $t \in Y(\nu S) \cap \mathcal{T}$, $\gamma = \exp_G(t)$, and an addition law $E^{(j)}$ of the embedding with $E^{(j)}(\xi(\gamma), \xi(0)) \neq 0$. For $z = \sum s_i w_i \in W$ small put $F_t(s) = Q_S(\lambda(\gamma) + \lambda(z), \mu(\gamma)\mu(z), E^{(j)}(\xi(\gamma), \xi(z)))$. By homogeneity, F_t is a unit times $Q_S(\lambda, \mu, \psi)(t+z)/\psi_0(t+z)^{D^\circ}$, so it has the same order along W. Its Taylor coefficients θ' of order $< T$ satisfy $|\theta'| \leq e^{-\nu^6\Delta + c\nu\Delta}$ (Cauchy). They are polynomials, with no denominator depending on γ, in the coefficients of Q_S, the coordinates of γ, and the Taylor coefficients of λ, μ, ξ along W at the origin. The latter are K_0-rational with house and denominator $\leq c^{|\tau|+1}(|\tau|!)^c$: by Bertrand's Corollaire to his Lemme 7 [25, App. I, p. 177] the coordinates of the exponential map have finite arithmetic functional order, every point being regular, and his Lemme 1 [25, App. I, pp. 166–167] turns this into such bounds for derivatives ([21, p. 396] cites this as "Lemma 7"). Since $|\tau| < T$ and $T \log T \leq c\Delta$ (as $a \geq 1$ and $\log \Delta \leq c \log S$), Serre's height bounds for γ then give $h(\theta') \leq c\nu\Delta$ with c independent of ν. Liouville's inequality forces $\theta' = 0$ once ν^5 exceeds a constant. (Dividing by a power of $E_0^{(j)}(\xi(\gamma), \xi(0))$ instead would cost a height $T(\nu S)^2$, which is too large.)

Step 3 (translation). Write $Y(S) \subset (Y(\nu S) \cap \mathcal{T}) + F'$ with F' finite. For $f \in F'$ take the product over the addition laws $E^{(j)}$ with $E^{(j)}(\cdot, \xi(-\exp f)) \not\equiv 0$ of Q_S translated by $-\exp f$; then take the product over $f \in F'$. Translations commute with invariant derivations, and each factor is a unit times the translate on a dense open set, so the product vanishes with multiplicity $\geq T$ along W on $\exp_G(Y(S))$. It is not identically zero on G, and its degrees are $\leq D_i$ once ν exceeds (number of laws)·(bidegree)·$|F'|$. No norm to $\mathbb{Q}$ is taken, since [21, Prop. 6.1] allows $\overline{\mathbb{Q}}$-coefficients.

The exponent $b \geq 1$ changes only the number of unknowns, which is compensated exactly by the factor $(\log S)^{bd_0}$ of Δ^{d-N}, and lowers the height term $(D_0/\nu)\log S$. In the proof of [21, Thm. 4.1], $b > 1$ is used only in the case $\delta = \tau$, which requires $d_0 \geq 1$ (Theorem 3). □

Remark 33 (Addition laws)**.** Let $G_2 \subset \mathbb{P}_M$ be locally closed over K_0, and $X = G_2 \times G_2 \subset \mathbb{P}_M \times \mathbb{P}_M$. The local ring of X at a point p is a quotient of that of $\mathbb{P}_M \times \mathbb{P}_M$, whose elements are ratios of bihomogeneous forms of equal bidegree. Clearing denominators, near p the group

law $X \to G_2 \subset \mathbb{P}_M$ is given by bihomogeneous forms $(A_0, \dots, A_M)$ of one bidegree, not all zero at p. By quasi-compactness finitely many such systems suffice, and multiplying each by suitable monomials $x_i^a y_l^b$ gives a complete system of addition laws of one bidegree (β, β), with coefficients in a finite extension of K_0. (This replaces K_0 by a finite extension in Step 2, which only changes $[K_0 : \mathbb{Q}]$ in the Liouville inequality.) Step 2 uses these laws only on $G_2(K_0) \times \exp(\text{a neighbourhood of 0 in } W)$, and Step 3 only with a fixed second argument.

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